4-3-4

1. The amount of algae, measured in micrograms, in a small pond depends on the temperature of the water in the pond and is modeled by the function $A(p) = \frac{3}{2}\left(2.7^{p^2/100}\right)$ for $0 \le p \le 30$, where p is the temperature of the water in the pond measured in degrees Celsius ($^\circ\mathrm{C}$). At the instant when the temperature is $20^\circ\mathrm{C}$, the rate of change of the temperature with respect to time is $1.5^\circ\mathrm{C}$ per day.

At what rate is the amount of algae changing, in micrograms per day, at that instant?

(A) 31.671

(B) 47.507

(C) 79.716

(D) 119.574

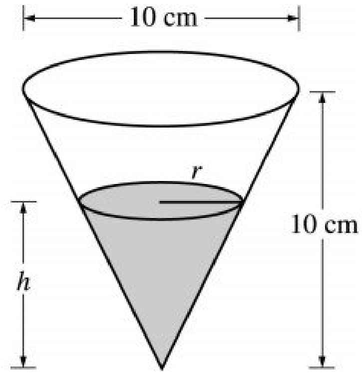
Answer B

Correct. The rate of change of the amount of algae with respect to time is $\frac{dA}{dt}$. By the chain rule, at the instant when $p = 20$, $\frac{dA}{dt} = \left(\frac{dA}{dp} \cdot \frac{dp}{dt}\right)\Big|_{p=20} = \left(A'(20)\,\frac{\text{mcg}}{^\circ\mathrm{C}}\right) \cdot \left(1.5\,\frac{^\circ\mathrm{C}}{\text{day}}\right) = 47.507\,\frac{\text{mcg}}{\text{day}}$.

The graphing calculator is used to compute the derivative value.

4-3-4

2.



A container has the shape of an open right circular cone, as shown in the figure above. The height of the container is 10 cm and the diameter of the opening is 10 cm. Water in the container is evaporating so that its depth h is changing at the constant rate of $\frac{-3}{10}$ cm/hr.

(Note: The volume of a cone of height h and radius r is given by $V = \frac{1}{3}\pi r^2 h$.)

(a) Find the volume V of water in the container when $h = 5$ cm. Indicate units of measure.

(b) Find the rate of change of the volume of water in the container, with respect to time, when $h = 5$ cm. Indicate units of measure.

(c) Show that the rate of change of the volume of water in the container due to evaporation is directly proportional to the exposed surface area of the water. What is the constant of proportionality?

Part A

1 point is earned for finding V when $h = 5$

When $h = 5$, $r = \frac{5}{2}$; $V(5) = \frac{1}{3}\pi\left(\frac{5}{2}\right)^2 5 = \frac{125}{12}\pi \text{ cm}^3$

0	1

The student response earns one of the following points:

1 point is earned for finding V when $h = 5$

When $h = 5$, $r = \frac{5}{2}$; $V(5) = \frac{1}{3}\pi\left(\frac{5}{2}\right)^2 5 = \frac{125}{12}\pi \text{ cm}^3$

Part B

4-3-4

1 point is earned for $r = \dfrac{1}{2}h$ in (a) or (b)

1 point is earned for V as a function of one variable in (a) or (b) OR $\dfrac{dr}{dt}$

2 points are earned for $\dfrac{dV}{dt}$

<-2> chain rule or product rule error

1 point is earned for the evaluation at $h = 5$

$$\frac{r}{h} = \frac{5}{10}, \text{ so } r = \frac{1}{2}h$$
$$V = \frac{1}{3}\pi\left(\frac{1}{4}h^2\right)h = \frac{1}{12}\pi h^3;\ \frac{dV}{dt} = \frac{1}{4}\pi h^2\frac{dh}{dt}$$
$$\left.\frac{dV}{dt}\right|_{h=5} = \frac{1}{4}\pi(25)\left(-\frac{3}{10}\right) = -\frac{15}{8}\pi \text{ cm}^3/\text{hr}$$

OR

$$\frac{dV}{dt} = \frac{1}{3}\pi\left(r^2\frac{dh}{dt} + 2rh\frac{dr}{dt}\right);\ \frac{dr}{dt} = \frac{1}{2}\frac{dh}{dt}$$
$$\left.\frac{dV}{dt}\right|_{h=5,r=\frac{5}{2}} = \frac{1}{3}\pi\left(\left(\frac{25}{4}\right)\left(-\frac{3}{10}\right) + 2\left(\frac{5}{2}\right)5\left(-\frac{3}{20}\right)\right)$$
$$= -\frac{15}{8}\pi \text{ cm}^3/\text{hr}$$

0	1	2	3	4	5

The student response earns all of the following points:

1 point is earned for $r = \dfrac{1}{2}h$ in (a) or (b)

1 point is earned for V as a function of one variable in (a) or (b) OR $\dfrac{dr}{dt}$

2 points are earned for $\dfrac{dV}{dt}$

<-2> chain rule or product rule error

1 point is earned for the evaluation at $h = 5$

4-3-4

$\frac{r}{h} = \frac{5}{10}$, so $r = \frac{1}{2}h$

$V = \frac{1}{3}\pi\left(\frac{1}{4}h^2\right)h = \frac{1}{12}\pi h^3;\ \frac{dV}{dt} = \frac{1}{4}\pi h^2 \frac{dh}{dt}$

$\frac{dV}{dt}\Big|_{h=5} = \frac{1}{4}\pi(25)\left(-\frac{3}{10}\right) = -\frac{15}{8}\pi\ \text{cm}^3/\text{hr}$

OR

$\frac{dV}{dt} = \frac{1}{3}\pi\left(r^2\frac{dh}{dt} + 2rh\frac{dr}{dt}\right);\ \frac{dr}{dt} = \frac{1}{2}\frac{dh}{dt}$

$$\frac{dV}{dt}\Big|_{h=5, r=\frac{5}{2}} = \frac{1}{3}\pi\left(\left(\frac{25}{4}\right)\left(-\frac{3}{10}\right) + 2\left(\frac{5}{2}\right)5\left(-\frac{3}{20}\right)\right)$$
$$= -\frac{15}{8}\pi\ \text{cm}^3/\text{hr}$$

Part C

1 point is earned for showing $\frac{dV}{dt} = k \cdot \text{area}$

1 point is earned for identifying constant of proportionality

$$\frac{dV}{dt} = \frac{1}{4}\pi h^2\frac{dh}{dt} = -\frac{3}{40}\pi h^2$$
$$= -\frac{3}{40}\pi(2r)^2 = -\frac{3}{10}\pi r^2 = -\frac{3}{10} \cdot \text{area}$$

The constant of proportionality is $-\frac{3}{10}$.

0	1	2

The student response earns all of the following points:

1 point is earned for showing $\frac{dV}{dt} = k \cdot \text{area}$

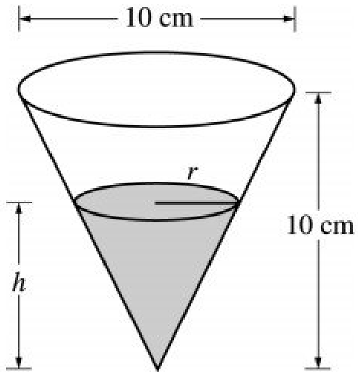
1 point is earned for identifying constant of proportionality

4-3-4

$$\frac{dV}{dt} = \tfrac{1}{4}\pi h^2 \frac{dh}{dt} = -\frac{3}{40}\pi h^2$$
$$= -\frac{3}{40}\pi(2r)^2 = -\frac{3}{10}\pi r^2 = -\frac{3}{10}\cdot \text{area}$$

The constant of proportionality is $-\dfrac{3}{10}$.

4-3-4



A container has the shape of an open right circular cone, as shown in the figure above. The height of the contained is 10 cm and the diameter of the opening is 10 cm. Water in the contained is evaporating so that its depth h is changing at the constant rate of $\frac{-3}{10}$ cm/hr.

3. Find the rate of change of the volume of water in the contained, with respect to time, when $h = 5$ cm. Indicate units of measure.

Part B

1 point is earned for $r = \frac{1}{2}h, in(a)or(b)$

1 point is earned for V as a function of one variable in (a) or (b)

or $\frac{dr}{dt}$

2 points are earned for $\frac{dV}{dt}$

<-2> chain rule or product rule error

1 point is earned for the evaluation at $h = 5$

$$\frac{r}{h} = \frac{5}{10}, so, r = \frac{1}{2}h$$
$$V = \frac{1}{3}\pi\left(\frac{1}{4}h^2\right)h = \frac{1}{12}\pi h^3; \frac{dV}{dt} = \frac{1}{4}\pi h^2\frac{dh}{dt}$$
$$\frac{dV}{dt}\Big|_{h=5} = \frac{1}{4}\pi(25)\left(-\frac{3}{10}\right) = -\frac{15}{8}\pi, cm^3/hr$$
$$OR$$
$$\frac{dV}{dt} = \frac{1}{3}\pi\left(r^2\frac{dh}{dt} + 2rh\frac{dr}{dt}\right); \frac{dr}{dt} = \frac{1}{2}\frac{dh}{dt}$$
$$\frac{dV}{dt}\Big|_{h=5,r=\frac{5}{2}} = \frac{1}{3}\pi\left(\left|\left(\frac{25}{5}\right)\left(-\frac{3}{10}\right) + 2\left(\frac{5}{2}\right)5\left(-\frac{3}{20}\right)\right)$$
$$= -\frac{15}{8}\pi, cm^3/hr$$

4-3-4

0	1	2	3	4	5

The student response earns all of the following points:

1 point is earned for $r = \frac{1}{2}h, in(a) or(b)$

1 point is earned for V as a function of one variable in (a) or (b)

or $\frac{dr}{dt}$

2 points are earned for $\frac{dV}{dt}$

<-2> chain rule or product rule error

1 point is earned for the evaluation at $h = 5$

$$\frac{r}{h} = \frac{5}{10}, so, r = \frac{1}{2}h$$
$$V = \frac{1}{3}\pi\left(\frac{1}{4}h^2\right)h = \frac{1}{12}\pi h^3; \frac{dV}{dt} = \frac{1}{4}\pi h^2 \frac{dh}{dt}$$
$$\left.\frac{dV}{dt}\right|_{h=5} = \frac{1}{4}\pi(25)\left(-\frac{3}{10}\right) = -\frac{15}{8}\pi, cm^3/hr$$
$$OR$$
$$\frac{dV}{dt} = \frac{1}{3}\pi\left(r^2\frac{dh}{dt} + 2rh\frac{dr}{dt}\right); \frac{dr}{dt} = \frac{1}{2}\frac{dh}{dt}$$
$$\left.\frac{dV}{dt}\right|_{h=5, r=\frac{5}{2}} = \frac{1}{3}\pi\left(\left|\left(\frac{25}{5}\right)\left(-\frac{3}{10}\right) + 2\left(\frac{5}{2}\right)5\left(-\frac{3}{20}\right)\right)\right.$$
$$= -\frac{15}{8}\pi, cm^3/hr$$

4-3-4

4. Show that the rate of change of the volume of water in the contained due to evaporation is directly proportional to the exposed surface area of the water. What is the constant of proportionality?

Part C

1 point is earned for shows $\frac{dV}{dt}$ = k area

1 point is earned for identifies constant of proportionality

$$\frac{dV}{dt} = \frac{1}{4}\pi h^2 \frac{dh}{dt} = -\frac{3}{40}\pi h^2$$
$$= -\frac{3}{40}\pi(2r)^2 = -\frac{3}{10}\pi r^2 = -\frac{3}{10} \cdot area$$

The constant of proportionality is $-\frac{3}{10}$.

0	1	2

The student response earns all of the following points:

1 point is earned for shows $\frac{dV}{dt}$ = k area

1 point is earned for identifies constant of proportionality

$$\frac{dV}{dt} = \frac{1}{4}\pi h^2 \frac{dh}{dt} = -\frac{3}{40}\pi h^2$$
$$= -\frac{3}{40}\pi(2r)^2 = -\frac{3}{10}\pi r^2 = -\frac{3}{10} \cdot area$$

The constant of proportionality is $-\frac{3}{10}$.

4-3-4

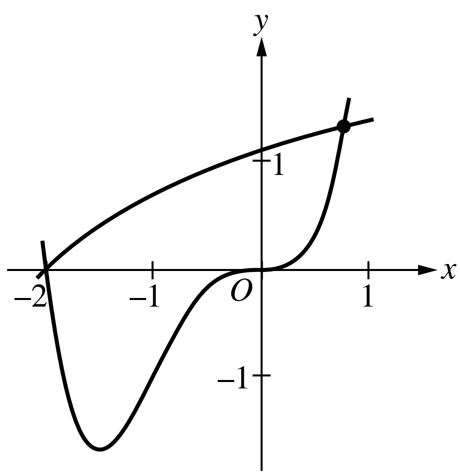
5. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Let f and g be the functions defined by $f(x) = \ln(x+3)$ and $g(x) = x^4 + 2x^3$. The graphs of f and g, shown in the figure, intersect at $x = -2$ and $x = B$, where $B > 0$.

(a) Find the area of the region enclosed by the graphs of f and g.

(b) For $-2 \le x \le B$, let $h(x)$ be the vertical distance between the graphs of f and g. Is h increasing or decreasing at $x = -0.5$? Give a reason for your answer.

(c) The region enclosed by the graphs of f and g is the base of a solid. Cross sections of the solid taken perpendicular to the x-axis are squares. Find the volume of the solid.

(d) A vertical line in the xy-plane travels from left to right along the base of the solid described in part (c). The vertical line is moving at a constant rate of 7 units per second. Find the rate of change of the area of the cross section above the vertical line with respect to time when the vertical line is at position $x = -0.5$.

Part A

4-3-4

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Other forms of the integrand in a definite integral, e.g., $|f(x) - g(x)|$, $|g(x) - f(x)|$, or $g(x) - f(x)$, earn the first point.

To earn the second point, the response must have a lower limit of -2 and an upper limit expressed as either the letter B with no value attached, or a number that is correct to the number of digits presented, with at least one and up to three decimal places.

· Case 1: If the response did not earn the second point because of an incorrect value of B, $0 < B < 1$, but used a lower limit of -2, the response earns the third point only for a consistent answer.

· Case 2: If the response did not earn the second point because the lower limit used was $x = 0$, but the response used a correct upper limit of B the response earns the third point for a consistent answer of 0.708 (or 0.707).

· Case 3: If a response uses any other incorrect limits it does not earn the second or third points.

A response containing the integrand $g(x) - f(x)$ must interpret the value of the resulting integral correctly to earn the third point. For example, the following response earns all 3 points: $\int_{-2}^{B} (g(x) - f(x))\, dx = -3.604$ so the area is 3.604. However, the response "Area $= \int_{-2}^{B} (g(x) - f(x))\, dx = 3.604$" presents an untrue statement and earns the first and second points but not the third point.

A response must earn the first point in order to be eligible for the third point. If the response has earned the second point, then only the correct answer will earn the third point.

Instructions for scoring a response that presents an integrand of $\ln(x+3) - x^4 + 2x^3$:

· The first time a response implicitly presents $f(x) - g(x)$ as $j(x) = \ln(x+3) - x^4 + 2x^3$ (with no explicit connection) in any part of this question, the response loses a point. The response is then eligible for all remaining points for a correct or consistent answer.

· In part (a) the consistent answer using $j(x)$ is negative and will not earn the third point.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- ☐ Integrand
- ☐ Limits of integration
- ☐ Answer

4-3-4

Solution:

$\ln(x+3) = x^4 + 2x^3 \;\Rightarrow\; x = -2,\;\; x = B = 0.781975$

$$\int_{-2}^{B} (f(x) - g(x))\, dx = 3.603548$$

The area of the region is 3.604 (or 3.603).

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response need not present the value of $h'(-0.5)$. The last line earns both points. However, if a value is presented it must be correct for the digits reported up to three decimal places.

A response that reports an incorrect value of $h'(-0.5)$ earns only the first point.

A response that presents only $h'(x)$ does not earn either point.

A response that compares the values of $f'(x)$ and $g'(x)$ at $x = -0.5$ earns the first point and is eligible for the second point. This comparison can be made symbolically or verbally; for example, the response "the rate of change of $f(x)$ is less than the rate of change of $g(x)$ at $x = -0.5$" earns the first point.

Instructions for scoring a response that presents $h(x)$ as $\ln(x+3) - x^4 + 2x^3$:

· If the first time the response presents $f(x) - g(x)$ as $j(x) = \ln(x+3) - x^4 + 2x^3$ occurs in part (b), the response loses a point. If the response already lost a point for this error in part (a), then the response is eligible for all remaining points for a correct or consistent answer.

· In part (b), the consistent answer using $j(x)$ is that $j'(-0.5) = 2.4 > 0$, so h is increasing at $x = -0.5$.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ Considers $h'(-0.5)$ – OR – $f'(x) - g'(x)$
- ☐ Answer with reason

Solution:

$h(x) = f(x) - g(x)$

$h'(x) = f'(x) - g'(x)$

4-3-4

$h'(-0.5) = f'(-0.5) - g'(-0.5) = -0.6$ (or -0.599)

Since $h'(-0.5) < 0$, h is decreasing at $x = -0.5$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for an integrand of $k(f(x) - g(x))^2$ or its equivalent with $k \neq 0$ in any definite integral. If $k \neq 1$, then the response is not eligible for the second point.

A response that does not earn the first point is ineligible to earn the second point, with the following exceptions:

· A response which has a presentation error in the integrand (for example, mismatched or missing parentheses, misplaced exponent) does not earn the first point but would earn the second point for the correct answer. A response which has a presentation error in the integrand and which reports an incorrect answer earns no points.

· A response that presents an integrand of $\left(\ln(x+3) - x^4 + 2x^3\right)^2$.

If the first time the response presents $f(x) - g(x)$ as $j(x) = \ln(x+3) - x^4 + 2x^3$ occurs in part (c), the response loses a point. If the response already lost a point for this error in part (a) or part (b), then the response is eligible for all remaining points for a correct or consistent answer.

In part (c), the consistent answer using $j(x)$ is 252.187 252.187 (or 252.188).

A response that uses incorrect limits is only eligible for the second point, provided the limits are imported from part (a) in Case 1 or Case 2. In both of these situations, the second point is earned only for answers consistent with the imported limits.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Integrand
- ☐ Answer

Solution:

$$\int_{-2}^{B} (f(x) - g(x))^2\, dx = 5.340102$$

The volume of the solid is 5.340.

Part D

4-3-4

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point may be earned by presenting $\frac{dA}{dx} \cdot \frac{dx}{dt}$, $A' \cdot \frac{dx}{dt}$, $A'(x) \cdot \frac{dx}{dt}$, $A'(x) \cdot x'$, $A'(-0.5) \cdot 7$, or $k \cdot 7$, where k is a declared value of $A'(-0.5)$, or any equivalent expression, including $2(f(x) - g(x))(f'(x) - g'(x))\frac{dx}{dt}$.

If a response defines $f(x) - g(x)$ as a function in parts (b) or (c) (for example, $h(x) = f(x) - g(x)$), then a correct expression for $\frac{dA}{dt}$ (for example, $2h\frac{dh}{dt}$) earns the first point.

A response that imports a function $A(x)$ declared in part (c) is eligible for both points (the answer must be consistent with the imported function $A(x)$).

A response that presents an incorrect function for $A(x)$ that is not imported from part (c) is eligible only for the first point.

Except when $A(x)$ is imported from part (c), the second point is earned only for the correct answer.

A response that does not earn the first point is ineligible to earn the second point except in the special case noted below.

· If the first time the response presents $f(x) - g(x)$ as $j(x) = \ln(x+3) - x^4 + 2x^3$ occurs in part (d), the response loses a point. If the response already lost a point for this error in part (a), part (b), or part (c), then the response is eligible for all remaining points for a correct or consistent answer. In part (d) the consistent answer using $j(x)$ is 20.287.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ $\frac{dA}{dx} \cdot \frac{dx}{dt}$
- ☐ Answer

Solution:

The cross section has area $A(x) = (f(x) - g(x))^2$.

$$\frac{d}{dt}[A(x)] = \frac{dA}{dx} \cdot \frac{dx}{dt}$$

$$\frac{d}{dt}[A(x)]\Big|_{x=-0.5} = A'(-0.5) \cdot 7 = -9.271842$$

At $x = -0.5$, the area of the cross section above the line is changing at a rate of -9.272 (or -9.271) square units per second.

4-3-4

6. The rate at which a baby bird gains weight is proportional to the difference between its adult weight and its current weight. At time $t = 0$, when the bird is first weighed, its weight is 20 grams. If $B(t)$ is the weight of the bird, in grams, at time t days after it is first weighed, then $\frac{db}{dt} = \frac{1}{5}(100 - B)$

Let $y = B(t)$ be the solution to the differential equation above with initial condition $B(0) = 20$.

Is the bird gaining weight faster when it weighs 40 grams or when it weighs 70 grams? Explain your reasoning.

Part A

One point is earned for the uses

One point is earned for the answer with reason

$$\frac{dB}{dt}|_{B=40} = \frac{1}{5}(60) = 12$$
$$\frac{dB}{dt}|_{B=70} = \frac{1}{5}(30) = 6$$

Because $\frac{dB}{dt}|_{B=40} > \frac{dB}{dt}|_{B=70}$, the bird is gaining weight faster when it weighs 40 grams.

0	1	2

The student earns all of the following points:

One point is earned for the uses

One point is earned for the answer with reason

$$\frac{dB}{dt}|_{B=40} = \frac{1}{5}(60) = 12$$
$$\frac{dB}{dt}|_{B=70} = \frac{1}{5}(30) = 6$$

Because $\frac{dB}{dt}|_{B=40} > \frac{dB}{dt}|_{B=70}$, the bird is gaining weight faster when it weighs 40 grams.

4-3-4

7. A 12,000-liter tank of water is filled to capacity. At time $t = 0$, water begins to drain out of the tank at a rate modeled by $r(t)$, measured in liters per hour, where r is given by the piecewise-defined function

$$r(t) = \begin{cases} \frac{600t}{t+3} & \text{for } 0 \le t \le 5 \\ 1000e^{-0.2t} & \text{for } t > 5 \end{cases}$$

Find $r'(3)$. Using correct units, explain the meaning of that value in the context of this problem.

Part C

1 point is earned for $r'(3)$

1 point is earned for meaning of $r'(3)$

r'(3)=50 The rate at which water is draining out of the tank at time t=3 hours is increasing at 50 liters/ hour^2.

0	1	2

The student response earns all of the following points:

1 point is earned for $r'(3)$

1 point is earned for meaning of $r'(3)$

r'(3)=50

The rate at which water is draining out of the tank at time t=3 hours is increasing at 50 liters/ hour^2.

4-3-4

8. The penguin population on an island is modeled by a differentiable function P of time t, where $P(t)$ is the number of penguins and t is measured in years for $0 \le t \le 40$. There are 100,000 penguins on the island at time $t = 0$. The birth rate for the penguins on the island is modeled by

$B(t) = 1000e^{0.06t}$ penguins per year

and the death rate for the penguins on the island is modeled by

$D(t)=250e^{0.1t}$ penguins per year.

What is the rate of change for the penguin population on the island at time $t = 0$?

Part A

The response can earn up to 1 point:

1 point: For the correct answer

$P'(0) = B(0)-D(0) = 1000 - 250 = 750$ penguins per year

0	1 ✓

The response earns the following point:

1 point: For the correct answer

$P'(0) = B(0)-D(0) = 1000 - 250 = 750$ penguins per year

9.

t (hours)	0	1	2	3	4	5	6
$s(t)$ (miles)	0	25	55	92	150	210	275

The table above gives the distance $s(t)$, in miles, that a car has traveled at various times t, in hours, during a 6-hour trip. The graph of the function s is increasing and concave up. Based on the information, which of the following could be the velocity of the car, in miles per hour, at time $t = 3$?

(A) 37
(B) 49 ✓
(C) 58
(D) 65
(E) 92

4-3-4

For $0 \le t \le 31$, the rate of change of the number of mosquitoes on Tropical Island at time *t* days is modeled by $R(t) = 5\sqrt{t}\cos\left(\frac{t}{5}\right)$ mosquitoes per day. There are 1000 mosquitoes on Tropical Island at time *t*=0.

10. At time *t*=6, is the number of mosquitoes increasing at an increasing rate, or is the number of mosquitoes increasing at a decreasing rate? Give a reason for your answer.

Part B

1 point is earned for considering R'(6)

1 point is earned for the answer with reason

R'(6)=-1.913 Since *R*'(6)<0, the number of mosquitoes is increasing at a decreasing rate at *t*=6.

0	1	2

The student response earns all of the following points:

1 point is earned for considering R'(6)

1 point is earned for the answer with reason

R'(6)=-1.913 Since *R*'(6)<0, the number of mosquitoes is increasing at a decreasing rate at *t*=6.

11. Show that the number of mosquitoes is increasing at time *t*=6.

Part A

1 point is earned for showing that $R(6) > 0$

Since *R*(6)=4.438>0, the number of mosquitoes is increasing at *t*=6.

0	1

The student response earns one of the following points:

1 point is earned for showing that $R(6) > 0$

Since *R*(6)=4.438>0, the number of mosquitoes is increasing at *t*=6.

4-3-4

12.

t (hours)	0	2	5	7	8
$E(t)$ (hundreds of entries)	0	4	13	21	23

A zoo sponsored a one-day contest to name a new baby elephant. Zoo visitors deposited entries in a special box between noon (t=0) and 8 P.M. (t=8) . The number of entries in the box t hours after noon is modeled by a differentiable function E for $0 \le t \le 8$ Values of $E(t)$, in hundreds of entries, at various times t are shown in the table above.

According to the model from part (c), at what time were the entries being processed most quickly? Justify your answer.

Part D

1 point is earned for considering $P'(t)=0$

1 point is earned for identifying candidates

1 point is earned for answering with justification

P'(t)=0 when t=9.183503 and t=10.816497

t	$P(t)$
8	0
9.183503	5.088662
10.816497	2.911338
12	8

Entries are being processed most quickly at time t=12

0	1	2	3

The student response earns all of the following points:

1 point is earned for considering $P'(t)=0$

1 point is earned for identifying candidates

1 point is earned for answering with justification

P'(t)=0 when t=9.183503 and t=10.816497

4-3-4

t	$P(t)$
8	0
9.183503	5.088662
10.816497	2.911338
12	8

Entries are being processed most quickly at time t=12

13. The following are related to this scenario:

A store is having a 12-hour sale. The total number of shoppers who have entered the store t hours after the sale begins is modeled by the function S defined by $S(t)=0.5t^4-16t^3+144t^2$ for $0 \le t \le 12$. At time $t=0$, when the sale begins, there are no shoppers in the store.

At what rate are shoppers entering the store 3 hours after the start of the sale?

Part A

The response can earn up to 1 point:

1 point: For the correct answer

$S'(3) = 486$ shoppers/hour

0	1

The response earns the following point:

1 point: For the correct answer

$S'(3) = 486$ shoppers/hour

4-3-4

14. The tide removes sand from Sandy Point Beach at a rate modeled by the function R, given by $R(t) = 2 + 5\sin\left(\frac{4\pi t}{25}\right)$.

A pumping station adds sand to the beach at a rate modeled by the function S, given by $S(t) = \frac{15t}{1+3t}$.

Both $R(t)$ and $S(t)$ have units of cubic yards per hour and t is measured in hours for $0 \leq t \leq 6$. At time $t = 0$, the beach contains 2500 cubic yards of sand.

Find the rate at which the total amount of sand on the beach is changing at time $t = 4$.

Part C

1 point is earned for the answer

$$Y'(t) = S(t) - R(t)$$
$$Y'(4) = S(4) - R(4) = -1.908 or - 1.909 yd^3/hr$$

0	1

The student response earns all of the following points:

1 point is earned for the answer

$$Y'(t) = S(t) - R(t)$$
$$Y'(4) = S(4) - R(4) = -1.908 or - 1.909 yd^3/hr$$

4-3-4

15. On a certain workday, the rate, in tons per hour, at which unprocessed gravel arrives at a gravel processing plant is modeled by $G(t) = 90 + 45\cos(t^2/18)$, where t is measured in hours and $0 \leq t \leq 8$. At the beginning of the workday ($t = 0$), the plant has 500 tons of unprocessed gravel. During the hours of operation, $0 \leq t \leq 8$, the plant processes gravel at a constant rate of 100 tons per hour.

Is the amount of unprocessed gravel at the plant increasing or decreasing at time $t = 5$ hours? Show the work that leads to your answer.

Part C

1 point is earned for comparing $G(5)$ to 100

1 point is earned for the conclusion

$G(5)=98.140764<100$

At time t=5, the rate at which unprocessed gravel is arriving is less than the rate at which it is being processed. Therefore, the amount of unprocessed gravel at the plant is decreasing at time t=5.

0	1	2

The student response earns all of the following points:

1 point is earned for comparing $G(5)$ to 100

1 point is earned for the conclusion

$G(5)=98.140764<100$

At time t=5, the rate at which unprocessed gravel is arriving is less than the rate at which it is being processed. Therefore, the amount of unprocessed gravel at the plant is decreasing at time t=5.

4-3-4

16. A cylindrical can of radius 10 millimeters is used to measure rainfall in Stormville. The can is initially empty, and rain enters the can during a 60-day period. The height of water in the can is modeled by the function S , where $S(t)$ is measured in millimeters and t is measured in days for $0 \leq t \leq 60$. The rate at which the height of the water is rising in the can is given by $S'(t) = 2\sin(0.03t) + 1.5$.

During the same 60-day period, rain on Monsoon Mountain accumulates in a can identical to the one in Stormville. The height of the water in the can on Monsoon Mountain is modeled by the function M, where $M(t) = 1/400\,(3t^3 - 30t^2 + 330t)$. The height $M(t)$ is measured in millimeters, and t is measured in days for $0 \leq t \leq 60$. Let $D(t) = M'(t) - S'(t)$. Apply the Intermediate Value Theorem to the function D on the interval $0 \leq t \leq 60$ to justify that there exists a time t, $0 < t < 60$, at which the heights of water in the two cans are changing at the same rate.

Part D

1 point is earned if considers $D(0)$ and $D(60)$

1 point is earned for justification

$D(0) = -0.675 < 0$ and $D(60) = 69.37730 > 0$

Because D is continuous, the Intermediate Value Theorem implies that there is a time t, $0 < t < 60$, at which $D(t) = 0$. At this time, the heights of water in the two cans are changing at the same rate.

0	1	2 ✓

The student response earns all of the following points:

1 point is earned if considers $D(0)$ and $D(60)$

1 point is earned for justification

$D(0) = -0.675 < 0$ and $D(60) = 69.37730 > 0$

Because D is continuous, the Intermediate Value Theorem implies that there is a time t, $0 < t < 60$, at which $D(t) = 0$. At this time, the heights of water in the two cans are changing at the same rate.

4-3-4

17. The rate at which people enter an auditorium for a rock concert is modeled by the function R given by $R(t) = 1380t^2 - 675t^3$ for $0 \leq t \leq 2$ hours; $R(t)$ is measured in people per hour. No one is in the auditorium at time $t = 0$, when the doors open. The doors close and the concert begins at time $t = 2$.

Find the time when the rate at which people enter the auditorium is a maximum. Justify your answer.

Part B

1 point is earned if considers R'(t)=0

1 point is earned for interior critical point

1 point is earned for the answer plus justification

$R'(t) = 0$ when $t = 0$ and $t = 1.36296$ The maximum rate may occur at 0, $a = 1.36296$, or 2.

$R(0)=0$

$R(a) = 854.527$

$R(2)=120$

The maximum rate occurs when $t=1.362$ or $t=1.363$.

0	1	2	3

The student response earns three of the following points:

1 point is earned if considers R'(t)=0

1 point is earned for interior critical point

1 point is earned for the answer plus justification

$R'(t) = 0$ when $t = 0$ and $t = 1.36296$ The maximum rate may occur at 0, $a = 1.36296$, or 2.

$R(0)=0$

$R(a) = 854.527$

$R(2)=120$

The maximum rate occurs when $t=1.362$ or $t=1.363$.

4-3-4

18. For time $t \geq 0$ hours, let $r(t) = 120\left(1 - e^{-10t^2}\right)$ represent the speed, in kilometers per hour, at which a car travels along a straight road. The number of liters of gasoline used by the car to travel x kilometers is modeled by $g(x) = 0.05x(1 - e^{-x/2})$.

Find the rate of change with respect to time of the number of liters of gasoline used by the car when $t = 2$ hours. Indicate units of measure.

Part B

2 points are earned for using chain rule

1 point is earned for the answer with units

$$\frac{dg}{dt} = \frac{dg}{dx} \cdot \frac{dx}{dt}; \quad \frac{dx}{dt} = r(t)$$
$$\left.\frac{dg}{dt}\right|_{t=2} = \left.\frac{dg}{dx}\right|_{x=206.370} \cdot r(2)$$

=(0.050)(120)=6 liters/hour

0	1	2	3

The student response earns all of the following points:

2 points are earned for using chain rule

1 point is earned for the answer with units

$$\frac{dg}{dt} = \frac{dg}{dx} \cdot \frac{dx}{dt}; \quad \frac{dx}{dt} = r(t)$$
$$\left.\frac{dg}{dt}\right|_{t=2} = \left.\frac{dg}{dx}\right|_{x=206.370} \cdot r(2)$$

=(0.050)(120)=6 liters/hour

4-3-4

19. People are entering a building at a rate modeled by $f(t)$ people per hour and exiting the building at a rate modeled by $g(t)$ people per hour, where t is measured in hours. The functions f and g are nonnegative and differentiable for all times t. Which of the following inequalities indicates that the rate of change of the number of people in the building is increasing at time t ?

(A) $f(t) > 0$

(B) $f'(t) > 0$

(C) $f(t) - g(t) > 0$

(D) $f'(t) - g'(t) > 0$ ✓

Answer D

This option is correct. The expression $R(t) = f(t) - g(t)$ indicates the rate of change of the number of people in the building (rate in minus rate out). If the derivative $R'(t) = f'(t) - g'(t)$ is positive, then $R(t)$ is increasing, i.e. the rate of change of the number of people in the building is increasing.

20. If $P(t)$ is the size of a population at time t, which of the following differential equations describes linear growth in the size of the population?

(A) $\frac{dP}{dt} = 200$ ✓

(B) $\frac{dP}{dt} = 200t$

(C) $\frac{dP}{dt} = 100t^2$

(D) $\frac{dP}{dt} = 200P$

(E) $\frac{dP}{dt} = 200P^2$

4-3-4

21. Consider the curve given by $y^2=2+xy$.

Let x and y be functions of time t that are related by the equation $y^2 = 2 + xy$. At time $t=5$, the value of y is 3 and dy/dt=6 . Find the value of dx/dt at time $t=5$.

Part D

1 point is earned if solves for x

1 point is earned for the chain rule

1 point is earned for the answer

When y=3, $3^2 = 2 + 3x$ so $x=\frac{7}{3}$

$$\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt} = \frac{y}{2y-x} \cdot \frac{dx}{dt}$$

At $t = 5, 6 = \frac{3}{6-\frac{7}{8}} \cdot \frac{dx}{dt} = \frac{9}{11} \cdot \frac{dx}{dt}$

$$\frac{dx}{dt}\Big|_{t=5} = \frac{22}{3}$$

0	1	2	3

The student response earns all of the following points:

1 point is earned if solves for x

1 point is earned for the chain rule

1 point is earned for the answer

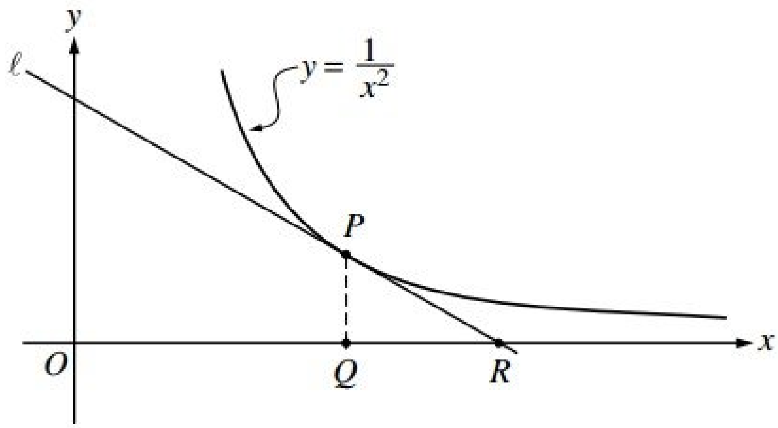
When y=3, $3^2 = 2 + 3x$ so $x=\frac{7}{3}$

$$\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt} = \frac{y}{2y-x} \cdot \frac{dx}{dt}$$

At $t = 5, 6 = \frac{3}{6-\frac{7}{8}} \cdot \frac{dx}{dt} = \frac{9}{11} \cdot \frac{dx}{dt}$

$$\frac{dx}{dt}\Big|_{t=5} = \frac{22}{3}$$

4-3-4



In the figure above, line L is tangent to the graph of $y = \frac{1}{x^2}$ at point P, with coordinates $\left(w, \frac{1}{w^2}\right)$, where $w > 0$. Point Q has coordinates $(w, 0)$. Line L crosses the x-axis at point R, with coordinates $(k, 0)$.

22. Suppose that w is increasing at the constant rate of 7 units per second. When $w = 5$, what is the rate of change of k with respect to time?

Part C

1 point is earned for the answer using: $\frac{dw}{dt} = 7$

$$\frac{dk}{dt} = \frac{3}{2}\frac{dw}{dt} = \frac{3}{2} \cdot 7 = \frac{21}{2}; \left.\frac{dk}{dt}\right|_{w=5} = \frac{21}{2}$$

0	1

The student response earns all of the following points:

1 point is earned for the answer using: $\frac{dw}{dt} = 7$

$$\frac{dk}{dt} = \frac{3}{2}\frac{dw}{dt} = \frac{3}{2} \cdot 7 = \frac{21}{2}; \left.\frac{dk}{dt}\right|_{w=5} = \frac{21}{2}$$

4-3-4

23. Suppose that w is increasing at the constant rate of 7 units per second. When $w = 5$, what is the rate of change of the area of nPQR with respect to time? Determine whether the area is increasing or decreasing at this instant.

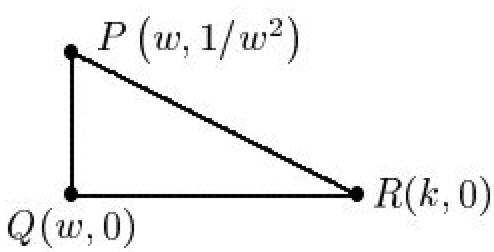
Part D

1 point is earned for the area in term or w and/or k

1 point is earned for: $\frac{dA}{dt}$ implicitly

1 point is earned for: $\left.\frac{dA}{dt}\right|_{w=5}$ using $\frac{dw}{dt} = 7$

1 point is earned for the conclusion



$$A = \tfrac{1}{2}(k - w)\frac{1}{w^2} = \tfrac{1}{2}\left(\tfrac{3}{2}w - w\right)\frac{1}{w^2} = \frac{1}{4w}$$
$$\frac{dA}{dt} = -\frac{1}{4w^2}\frac{dw}{dt}$$
$$\left.\frac{dA}{dt}\right|_{w=5} = -\frac{1}{100} \cdot 7 = -0.07$$

Therefore, area is decreasing.

0	1	2	3	4

The student response earns all of the following points:

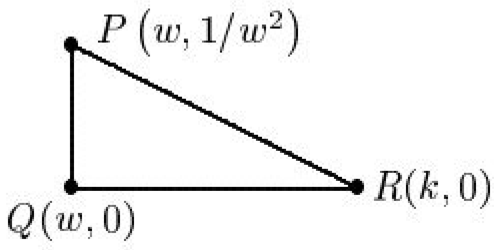
1 point is earned for the area in term or w and/or k

1 point is earned for: $\frac{dA}{dt}$ implicitly

4-3-4

1 point is earned for: $\frac{dA}{dt}\Big|_{w=5} u \sin g \frac{dw}{dt} - 7$

1 point is earned for the conclusion



$A = \frac{1}{2}(k-w)\frac{1}{w^2} = \frac{1}{2}\left(\frac{3}{2}w - w\right)\frac{1}{w^2} = \frac{1}{4w}$

$\frac{dA}{dt} = -\frac{1}{4w^2}\frac{dw}{dt}$

$\frac{dA}{dt}\Big|_{w=5} = -\frac{1}{100} \cdot 7 = -0.07$

Therefore, area is decreasing.

4-3-4

The radius r of a sphere is increasing at a constant rate of 0.04 centimeters per second. (Note: The volume of a sphere with radius r is $V=4/3\pi r^3$.)

24. At the time when the volume of the sphere is 36π cubic centimeters, what is the rate of increase of the area of a cross section through the center of the sphere?

Part B

1 point is earned for finding $r=3$

2 points are earned for derivative

1 point is **deducted** for incorrect area

1 point is **deducted** for each constant error

1 point is earned for x=3

$$V = 36\pi \Rightarrow 36 = \frac{4}{3}r^3 \Rightarrow r^3 = 27 \Rightarrow r = 3$$
$$A = \pi r^2$$
$$\frac{dA}{dt} = 2\pi r\frac{dr}{dt}$$
Therefore when $V = 36\pi, \frac{dr}{dt} = 0.04$
$$\frac{dA}{dt} = 2\pi \cdot 3(0.04) = \frac{6\pi}{25} = 0.24\pi \text{cm}^2/\text{ sec}$$

0	1	2	3	4

The student response earns four of the following points:

1 point is earned for finding $r=3$

2 points are earned for derivative

1 point is **deducted** for incorrect area

1 point is **deducted** for each constant error

1 point is earned for x=3

4-3-4

$V = 36\pi \Rightarrow 36 = \frac{4}{3}r^3 \Rightarrow r^3 = 27 \Rightarrow r = 3$

$A = \pi r^2$

$\frac{dA}{dt} = 2\pi r \frac{dr}{dt}$

Therefore when $V = 36\pi, \frac{dr}{dt} = 0.04$

$\frac{dA}{dt} = 2\pi \cdot 3(0.04) = \frac{6\pi}{25} = 0.24\pi \mathrm{cm}^2 / \sec$

4-3-4

25. At the time when the radius of the sphere is 10 centimeters, what is the rate of increase of its volume?

Part A

2 points are earned for derivative

2 points are **deducted** for chain rule error

1 point is deducted for constant error

1 point is earned for evaluation

$$\frac{dV}{dt} = \frac{4}{3} \cdot 3\pi r^2 \frac{dr}{dt}$$

Therefore when $r = 10$,

$$\frac{dr}{dt} = 0.04$$

$$\frac{dV}{dt} = 4\pi 10^2(0.04) = 16\pi \text{cm}^3/\text{ sec}$$

0	1	2	3

The student response earns three of the following points:

2 points are earned for derivative

2 points are **deducted** for chain rule error

1 point is deducted for constant error

1 point is earned for evaluation

$$\frac{dV}{dt} = \frac{4}{3} \cdot 3\pi r^2 \frac{dr}{dt}$$

Therefore when $r = 10$,

$$\frac{dr}{dt} = 0.04$$

$$\frac{dV}{dt} = 4\pi 10^2(0.04) = 16\pi \text{cm}^3/\text{ sec}$$

4-3-4

26. At the time when the volume and the radius of the sphere are increasing at the same numerical rate, what is the radius?

Part C

1 point is earned if sets $\frac{dV}{dt} = \frac{dr}{dt}$

1 point is earned for solution to equation

$$\frac{dV}{dt} = \frac{dr}{dt}$$

$$4\pi r^2 \frac{dr}{dt} = \frac{dr}{dt} \Rightarrow 4\pi r^2 = 1$$

$$\therefore r^2 \frac{1}{4\pi} \Rightarrow \boxed{r = \frac{1}{2\sqrt{\pi}}\, cm}$$

0	1	2

The student response earns none of the following points:

1 point is earned if sets $\frac{dV}{dt} = \frac{dr}{dt}$

1 point is earned for solution to equation

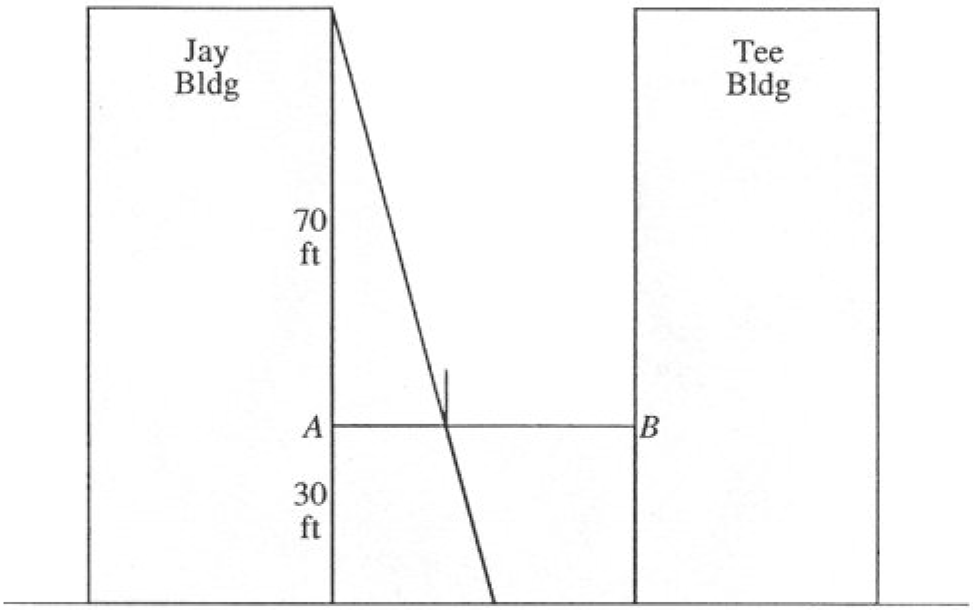
$$\frac{dV}{dt} = \frac{dr}{dt}$$

$$4\pi r^2 \frac{dr}{dt} = \frac{dr}{dt} \Rightarrow 4\pi r^2 = 1$$

$$\therefore r^2 \frac{1}{4\pi} \Rightarrow \boxed{r = \frac{1}{2\sqrt{\pi}}\, cm}$$

4-3-4

A tight rope is stretched 30 feet above the ground between the Jay and the Tee buildings, which are 50 feet apart. A tightrope walker, walking at a constant rate of 2 feet per second from point A to point B, is illuminated by a spotlight 70 feet above point A, as shown in the diagram.



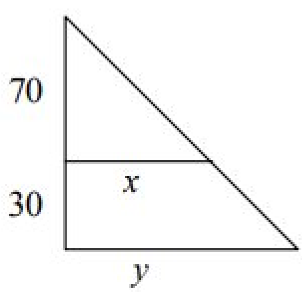
27. How fast is the shadow of the tightrope walker's feet moving along the ground when she is midway between the buildings? (Indicate units of measure.)

Part A

1 point is earned for establishing two-variable equation relating variables "x" and "y"

1 point is earned for establishing equation relating rates $\frac{dx}{dt}, \frac{dy}{dt}$

1 point is earned for finding answer



$$\frac{y}{100} = \frac{x}{70}$$

4-3-4

$$70\,\frac{dy}{dt} = 100\,(2)$$

$$\frac{dy}{dt} = \frac{10}{7}(2)\;\frac{20}{7}\ \mathrm{ft/sec}$$

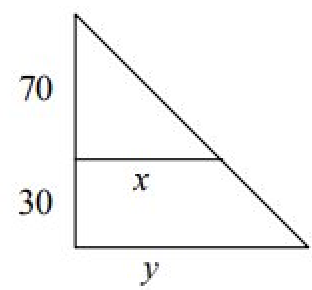
0	1	2	3

The student response earns three of the following points:

1 point is earned for establishing two-variable equation relating variables "x" and "y"

1 point is earned for establishing equation relating rates $\frac{dx}{dt}, \frac{dy}{dt}$

1 point is earned for finding answer



$$\frac{y}{100} = \frac{x}{70}$$

$$70\,\frac{dy}{dt} = 100\,(2)$$

$$\frac{dy}{dt} = \frac{10}{7}(2)\;\frac{20}{7}\ \mathrm{ft/sec}$$

4-3-4

28. How fast is the shadow of the tightrope walker's feet moving up the wall of the Tee Building when she is 10 feet from point *B*? (Indicate units of measure.)

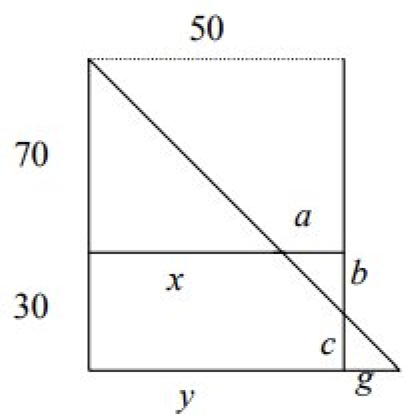
Part C

1 point is earned for establishing proportion relating horizontal variables x , a ,or g to vertical variables b or c

2 points are earned for establishing equation relating rates from student's proportion

Max 1/2 if student stops before getting an equation involving at most two variables and their rates

1 point is earned for answer (including units)



$\frac{y}{100} = \frac{x}{70}$

$70\frac{dy}{dt} = 100\frac{dx}{dt}\, OR \frac{dy/dt}{100} = \frac{dx/dt}{70}$

$\frac{b}{a} = \frac{70}{x}$

$\frac{b}{50-x} = \frac{70}{x}$

$bx = 3500 - 70x$

$b\frac{bx}{dt} + x\frac{db}{dt} = -70\frac{dx}{dt}$

$\left(\frac{35}{2}\right)(2) + (40)\frac{db}{dt} = -70\,(2)$

$\frac{db}{dt} = -\frac{35}{8}$

$\frac{35}{8}\mathrm{ft/\,sec}$

0	1	2	3	4

4-3-4

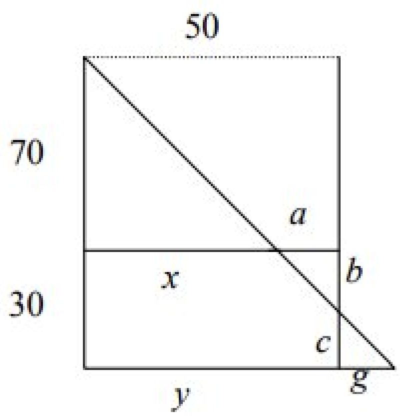
The student response earns four of the following points:

1 point is earned for establishing proportion relating horizontal variables x , a ,or g to vertical variables b or c

2 points are earned for establishing equation relating rates from student's proportion

Max 1/2 if student stops before getting an equation involving at most two variables and their rates

1 point is earned for answer (including units)



$\frac{y}{100} = \frac{x}{70}$

$70\frac{dy}{dt} = 100\frac{dx}{dt} OR \frac{dy/dt}{100} = \frac{dx/dt}{70}$

$\frac{b}{a} = \frac{70}{x}$

$\frac{b}{50-x} = \frac{70}{x}$

$bx = 3500 - 70x$

$b\frac{bx}{dt} + x\frac{db}{dt} = -70\frac{dx}{dt}$

$\left(\frac{35}{2}\right)(2) + (40)\frac{db}{dt} = -70\,(2)$

$\frac{db}{dt} = -\frac{35}{8}$

$\frac{35}{8}\text{ft}/\sec$

4-3-4

29. At time $t, t \geq 0$, the volume of a sphere is increasing at a rate proportional to the reciprocal of its radius. At $t = 0$, the radius of the sphere is 1 and at $t = 15$, the radius is 2. (The volume V of a sphere with a radius r is $V = 4/3\pi r^3$.)

Find the radius of the sphere as a function of t.

Part A

1 point is earned for $\frac{dV}{dt} = \frac{k}{r}$

1 point is earned for $\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$

1 point is earned for separates variables

1 point is earned for integration (must have C)

1 point is earned for solving for C

1 point is earned for solving for k

1 point is earned for answer

$$\frac{dV}{dt} = \frac{k}{r}$$
$$\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$$
$$\frac{k}{r} = 4\pi r^2 \frac{dr}{dt}$$
$$kdt = 4\pi r^3 dr$$
$$kt + C = \pi r^4$$
At $t = 0, r = 1$ so $C = \pi$

At $t = 15, r = 2$, so $15k + \pi = 16\pi, k = \pi$

$$\pi r^4 = \pi t + \pi$$
$$r = \sqrt[4]{t + 1}$$

0	1	2	3	4	5	6	7

The student response earns seven of the following points:

4-3-4

1 point is earned for $\dfrac{dV}{dt} = \dfrac{k}{r}$

1 point is earned for $\dfrac{dV}{dt} = 4\pi r^2 \dfrac{dr}{dt}$

1 point is earned for separates variables

1 point is earned for integration (must have C)

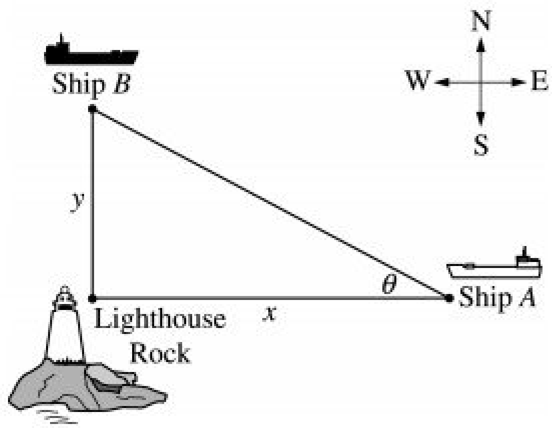
1 point is earned for solving for C

1 point is earned for solving for k

1 point is earned for answer

$\frac{dV}{dt} = \frac{k}{r}$
$\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$
$\frac{k}{r} = 4\pi r^2 \frac{dr}{dt}$
$kdt = 4\pi r^3 dr$
$kt + C = \pi r^4$
At $t = 0, r = 1$ so $C = \pi$
At $t = 15, r = 2,$ so $15k + \pi = 16\pi, k = \pi$
$\pi r^4 = \pi t + \pi$
$r = \sqrt[4]{t+1}$

4-3-4



Ship A is traveling due west toward Lighthouse Rock at a speed of 15 kilometers per hour (km/hr). Ship B is traveling due north away from Lighthouse rock at a speed of 10 km/hr. Let x be the distance between Ship A and Lighthouse Rock at time t, and let y be the distance between Ship B and Lighthouse Rock at time t, as shown in the figure above.

30. Find the rate of change, in km/hr, of the distance between the two ships when $x = 4$ km and $y = 3$ km.

Part B

1 point is earned for the expression for distance

2 points are earned for the differentiation with respect to t

< -2 > chain rule error

1 point is earned for evaluation

$$r^2 = x^2 + y^2$$
$$2r\frac{dr}{dt} = 2x\frac{dx}{dt} + 2y\frac{dy}{dt}$$

or explicitly:

$$r = \sqrt{x^2 + y^2}$$
$$\frac{dr}{dt} = \frac{1}{2\sqrt{x^2+y^2}}\left(2x\frac{dx}{dt} + 2y\frac{dy}{dt}\right)$$

At x = 4, y = 3,

$$\frac{dr}{dt} = \frac{4(-15)+3(10)}{5} = -6km/hr$$

4-3-4

0	1	2	3	4

The student response earns all of the following points:

1 point is earned for the expression for distance

2 points are earned for the differentiation with respect to *t*

< -2 > chain rule error

1 point is earned for evaluation

$$r^2 = x^2 + y^2$$
$$2r\frac{dr}{dt} = 2x\frac{dx}{dt} + 2y\frac{dy}{dt}$$

or explicitly:

$$r = \sqrt{x^2 + y^2}$$
$$\frac{dr}{dt} = \frac{1}{2\sqrt{x^2+y^2}}\left(2x\frac{dx}{dt} + 2y\frac{dy}{dt}\right)$$

At x = 4, y = 3,

$$\frac{dr}{dt} = \frac{4(-15)+3(10)}{5} = -6km/hr$$

4-3-4

31. Let θ be the angle shown in the figure. Find the rate of change of θ, radians per hour, when $x = 4$ km and $y = 3$ km.

Part C

1 point is earned for the expression for θ in terms of x and y

2 points are earned for differentiation with respect to t < -2 > chain rule, quotient rule, or transcendental function error

1 point is earned for the evaluation

$$\tan\theta = \frac{y}{x}$$

$$\sec^2\theta\frac{d\theta}{dt} = \frac{\frac{dy}{dt}x - \frac{dx}{dt}y}{x^2}$$

At x = 4 and y = 3, $\sec\theta = \frac{5}{4}$

$$\frac{d\theta}{dt} = \frac{16}{25}\left(\frac{10(4)-(-15)(3)}{16}\right)$$

$$= \frac{85}{25} = \frac{17}{5} radians/hr$$

0	1	2	3	4

The student response earns all of the following points:

1 point is earned for the expression for θ in terms of x and y

2 points are earned for differentiation with respect to t < -2 > chain rule, quotient rule, or transcendental function error

1 point is earned for the evaluation

$$\tan\theta = \frac{y}{x}$$

$$\sec^2\theta\frac{d\theta}{dt} = \frac{\frac{dy}{dt}x - \frac{dx}{dt}y}{x^2}$$

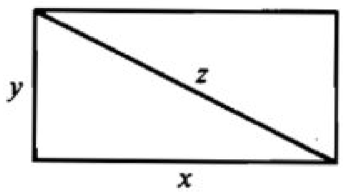
At x = 4 and y = 3, $\sec\theta = \frac{5}{4}$

4-3-4

$$\frac{d\theta}{dt} = \frac{16}{25}\left(\frac{10(4)-(-15)(3)}{16}\right)$$

$$= \frac{85}{25} = \frac{17}{5} radians/hr$$

32.



The sides of the rectangle above increase in such a way that $\frac{dz}{dt} = 1$ and $\frac{dx}{dt} = 3\frac{dy}{dt}$. At the instant when $x = 4$ and $y = 3$, what is the value of $\frac{dx}{dt}$?

(A) $\frac{1}{3}$

(B) 1 ✓

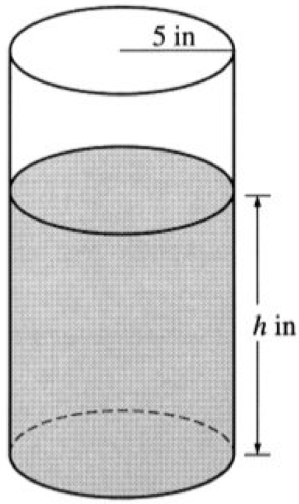
(C) 2

(D) $\sqrt{5}$

(E) 5

4-3-4

33.



A coffeepot has the shape of a cylinder with radius 5 inches, as shown in the figure above. Let h be the depth of the coffee in the pot, measured in inches, where h is a function of time t, measured in seconds. The volume V of coffee in the pot is changing at the rate of $-5\pi\sqrt{h}$ cubic inches per second. (The volume V of a cylinder with radius r and height h is $V = \pi r^2 h$.)

(a) Show that $\frac{dh}{dt} = -\frac{\sqrt{h}}{5}$.

(b) Given that $h = 17$ at time $t = 0$, solve the differential equation $\frac{dh}{dt} = -\frac{\sqrt{h}}{5}$ for h as a function of t.

(c) At what time t is the coffeepot empty?

Part A

1 point is earned for $\frac{dV}{dt} = -5\pi\sqrt{h}$

1 point is earned for computing $\frac{dV}{dt}$

1 point is earned for showing result

$$V = 25\pi h$$
$$\frac{dV}{dt} = 25\pi\frac{dh}{dt} = -5\pi\sqrt{h}$$
$$\frac{dh}{dt} = \frac{-5\pi\sqrt{h}}{25\pi} = -\frac{\sqrt{h}}{5}$$

0	1	2	3

The student response earns all of the following points:

1 point is earned for $\frac{dV}{dt} = -5\pi\sqrt{h}$

4-3-4

1 point is earned for computing $\frac{dV}{dt}$

1 point is earned for showing result

$$V = 25\pi h$$
$$\frac{dV}{dt} = 25\pi\frac{dh}{dt} = -5\pi\sqrt{h}$$
$$\frac{dh}{dt} = \frac{-5\pi\sqrt{h}}{25\pi} = -\frac{\sqrt{h}}{5}$$

Part B

1 point is earned for the correct separation of variables

$$\frac{dh}{dt} = -\frac{\sqrt{h}}{5}$$
$$\frac{1}{\sqrt{h}}dh = -\frac{1}{5}dt$$

1 point is earned for the correct antiderivatives

1 point is earned for the correct constant of integration

$$2\sqrt{h} = -\frac{1}{5}t + C$$

1 point is earned for using initial condition $h = 17$ when $t = 0$

$$2\sqrt{17} = 0 + C$$

1 point is earned for correctly solving for h

$$h = \left(-\frac{1}{10}t + \sqrt{17}\right)^2$$

Note: max 2/5 [1-1-0-0-0] if no constant of integration

Note: 0/5 if no separation of variables

0	1	2	3	4	5

The student response earns all of the following points:

1 point is earned for the correct separation of variables

$$\frac{dh}{dt} = -\frac{\sqrt{h}}{5}$$

4-3-4

$$\frac{1}{\sqrt{h}}dh = -\frac{1}{5}dt$$

1 point is earned for the correct antiderivatives

1 point is earned for the correct constant of integration

$$2\sqrt{h} = -\frac{1}{5}t + C$$

1 point is earned for using initial condition $h = 17$ when $t = 0$

$$2\sqrt{17} = 0 + C$$

1 point is earned for correctly solving for h

$$h = \left(-\frac{1}{10}t + \sqrt{17}\right)^2$$

Note: max 2/5 [1-1-0-0-0] if no constant of integration

Note: 0/5 if no separation of variables

Part C

1 point is earned for the correct answer

$$(-\frac{1}{10}t + \sqrt{17})^2 = 0$$

$$t = 10\sqrt{17}$$

0	1 ✓

The student response earns all of the following points:

1 point is earned for the correct answer

$$(-\frac{1}{10}t + \sqrt{17})^2 = 0$$

$$t = 10\sqrt{17}$$

4-3-4

34. The rate at which people enter an auditorium for a rock concert is modeled by the function R given by $R(t) = 1380t^2 - 675t^3$ for $0 \le t \le 2$ hours; $R(t)$ is measured in people per hour. No one is in the auditorium at time $t = 0$, when the doors open. The doors close and the concert begins at time $t = 2$.

(a) How many people are in the auditorium when the concert begins?

(b) Find the time when the rate at which people enter the auditorium is a maximum. Justify your answer.

(c) T*he total wait time for all the people in the auditorium is found by adding the time each person waits, starting at the time the person enters the auditorium and ending when the concert begins. The function w models the total wait time for all the people who enter the auditorium before time t*. The derivative of w is given by $w'(t) = (2 - t)R(t)$. Find $w(2) - w(1)$, the total wait time for those who enter the auditorium after time $t = 1$.

(d) On average, how long does a person wait in the auditorium for the concert to begin? Consider all people who enter the auditorium after the doors open, and use the model for total wait time from part (c).

Part A

1 point is earned for integral

1 point is earned for answer

$$\int_0^2 R(t)dt = 980 \text{ people}$$

0	1	2

The student response earns all of the following points:

1 point is earned for integral

1 point is earned for answer

$$\int_0^2 R(t)dt = 980 \text{ people}$$

Part B

1 point is earned if considers $R'(t) = 0$

1 point is earned for interior critical point

1 point is earned for the answer plus justification

4-3-4

$R'(t) = 0$ when $t = 0$ and $t = 1.36296$

The maximum rate may occur at 0, $a = 1.36296$, or 2.

$R(0) = 0$

$R(a) = 854.527$

$R(2) = 120$

The maximum rate occurs when $t = 1.362$ or 1.363.

0	1	2	3

The student response earns all of the following points:

1 point is earned if considers $R'(t) = 0$

1 point is earned for interior critical point

1 point is earned for the answer plus justification

$R'(t) = 0$ when $t = 0$ and $t = 1.36296$

The maximum rate may occur at 0, $a = 1.36296$, or 2.

$R(0) = 0$

$R(a) = 854.527$

$R(2) = 120$

The maximum rate occurs when $t = 1.362$ or 1.363.

Part C

1 point is earned for integral

1 point is earned for answer

$$w(2) - w(1) = \int_1^2 w'(t)dt = \int_1^2 (2-t)R(t)dt = 387.5$$

The total wait time for those who enter the auditorium after time $t = 1$ is 387.5 hours.

4-3-4

0	1	2

The student response earns all of the following points:

1 point is earned for integral

1 point is earned for answer

$$w(2) - w(1) = \int_1^2 w'(t)dt = \int_1^2 (2-t)R(t)dt = 387.5$$

The total wait time for those who enter the auditorium after time $t = 1$ is 387.5 hours.

Part D

1 point is earned for integral

1 point is earned for answer

$$\frac{1}{980}w(2) = \frac{1}{980}\int_0^2 (2-t)R(t)dt = 0.77551$$

On average, a person waits 0.775 or 0.776 hour.

0	1	2

The student response earns all of the following points:

1 point is earned for integral

1 point is earned for answer

$$\frac{1}{980}w(2) = \frac{1}{980}\int_0^2 (2-t)R(t)dt = 0.77551$$

On average, a person waits 0.775 or 0.776 hour.

4-3-4

35. For time $t \geq 0$ hours, let $r(t) = 120\left(1 - e^{-10t^2}\right)$ represent the speed, in kilometers per hour, at which a car travels along a straight road. The number of liters of gasoline used by the car to travel x kilometers is modeled by $g(x) = 0.05x(1 - e^{-x/2})$.

(a) How many kilometers does the car travel during the first 2 hours?

(b) Find the rate of change with respect to time of the number of liters of gasoline used by the car when $t = 2$ hours. Indicate units of measure.

(c) How many liters of gasoline have been used by the car when it reaches a speed of 80 kilometers per hour?

Part A

1 point is earned for the integral

1 point is earned for the answer

$$\int_0^2 r(t)\,dt = 206.370 \text{ kilometers}$$

0	1	2

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the answer

$$\int_0^2 r(t)\,dt = 206.370 \text{ kilometers}$$

Part B

2 points are earned for using chain rule

1 point is earned for the answer with units

$$\frac{dg}{dt} = \frac{dg}{dx} \cdot \frac{dx}{dt};\ \frac{dx}{dt} = r(t)$$
$$\left.\frac{dg}{dt}\right|_{t=2} = \left.\frac{dg}{dx}\right|_{x=206.370} \cdot r(2)$$

4-3-4

= (0.050)(120) = 6 liters/hour

0	1	2	3

The student response earns all of the following points:

2 points are earned for using chain rule

1 point is earned for the answer with units

$$\frac{dg}{dt} = \frac{dg}{dx} \cdot \frac{dx}{dt}; \ \frac{dx}{dt} = r(t)$$
$$\left.\frac{dg}{dt}\right|_{t=2} = \left.\frac{dg}{dx}\right|_{x=206.370} \cdot r(2)$$

= (0.050)(120) = 6 liters/hour

Part C

1 point is earned for equation $r(t) = 80$

2 points are earned for distance integral

1 point is earned for answer

Let T be the time at which the car's speed reaches 80 kilometers per hour.

Then, $r(T) = 80$ or $T = 0.331453$ hours.

At time T, the car has gone

$$x(T) = \int_0^T r(t)\, dt = 10.794097 \text{ kilometers}$$

and has consumed $g(x(T)) = 0.537$ liters of gasoline.

0	1	2	3	4

The student response earns all of the following points:

1 point is earned for equation $r(t) = 80$

2 points are earned for distance integral

4-3-4

1 point is earned for answer

Let T be the time at which the car's speed reaches 80 kilometers per hour.

Then, $r(T) = 80$ or $T = 0.331453$ hours.

At time T, the car has gone

$$x(T) = \int_0^T r(t)\, dt = 10.794097 \text{ kilometers}$$

and has consumed $g(x(T)) = 0.537$ liters of gasoline.

4-3-4

36. A store is having a 12-hour sale. The total number of shoppers who have entered the store t hours after the sale begins is modeled by the function S defined by $S(t) = 0.5t^4 - 16t^3 + 144t^2$ for $0 \le t \le 12$. At time $t = 0$, when the sale begins, there are no shoppers in the store.

(a) At what rate are shoppers entering the store 3 hours after the start of the sale?

(b) Find the value of $\frac{1}{3}\int_6^9 S'(t)\,dt$. Using correct units, explain the meaning of $\frac{1}{3}\int_6^9 S'(t)\,dt$ in the context of this problem.

(c) The rate at which shoppers leave the store, measured in shoppers per hour, is modeled by the function L defined by $L(t) = -80 + \frac{4400}{t^2 - 14t + 55}$ for $0 \le t \le 12$. According to the model, how many shoppers are in the store at the end of the sale (time $t = 12$)? Give your answer to the nearest whole number.

(d) Using the given models, find the time t, $0 \le t \le 12$, at which the number of shoppers in the store is the greatest. Justify your answer.

Part A

The response can earn up to 1 point:

1 point is earned for the answer

$S'(3) = 486$ shoppers/hour

0	1 ✓

The student earns all of the following point:

1 point is earned for the answer

$S'(3) = 486$ shoppers/hour

Part B

The response can earn up to 2 points:

1 point is earned for the value of $\frac{1}{3}\int_6^9 S'(t)\,dt$

$$\frac{1}{3}\int_6^9 S'(t)\,dt = \frac{1}{3}(S(9) - S(6))$$

$$= \frac{1}{3}(3280.5 - 2376) = 301.5$$

1 point is earned for the meaning of $\frac{1}{3}\int_6^9 S'(t)\,dt$

4-3-4

$\frac{1}{3}\int_{6}^{9} S'(t)\, dt$ is the average rate at which shoppers are entering the store in shoppers/hr between times $t = 6$ and $t = 9$ hours.

0	1	2

The student earns all of the following points:

1 point is earned for the value of $\frac{1}{3}\int_{6}^{9} S'(t)\, dt$

$$\frac{1}{3}\int_{6}^{9} S'(t)\, dt = \frac{1}{3}(S(9) - S(6))$$
$$= \frac{1}{3}(3280.5 - 2376) = 301.5$$

1 point is earned for the meaning of $\frac{1}{3}\int_{6}^{9} S'(t)\, dt$

$\frac{1}{3}\int_{6}^{9} S'(t)\, dt$ is the average rate at which shoppers are entering the store in shoppers/hr between times $t = 6$ and $t = 9$ hours.

Part C

The response can earn up to 3 points:

1 point is earned for the integral

1 point is for use of $S(12)$

1 point is for the answer

$$S(12) - \int_{0}^{12} L(t)\, dt = 195.701684$$

Therefore, there are approximately 196 shoppers in the store at the end of the sale.

0	1	2	3

The student earns all of the following points:

1 point is earned for the integral

4-3-4

1 point is for use of $S(12)$

1 point is for the answer

$$S(12) - \int_0^{12} L(t)\,dt = 195.701684$$

Therefore, there are approximately 196 shoppers in the store at the end of the sale.

Part D

The response can earn up to 3 points:

1 point is earned for considering $S'(t) - L(t) = 0$

1 point is earned for the answer

1 point is earned for justification

The number of shoppers in the store at time t is given by the function N defined by

$$N(t) = S(t) - \int_0^{t} L(x)\,dx$$

$N'(t) = S'(t) - L(t) = 0$ when $t = 5.545066$

t	$N(t)$
0	0
5.545066	1361.832842
12	195.702

The number of shoppers in the store is greatest at time $t = 5.545$ hours.

0	1	2	3

The student earns all of the following points:

1 point is earned for considering $S'(t) - L(t) = 0$

1 point is earned for the answer

1 point is earned for justification

The number of shoppers in the store at time t is given by the function N defined by

4-3-4

$$N(t) = S(t) - \int_0^t L(x)\,dx$$

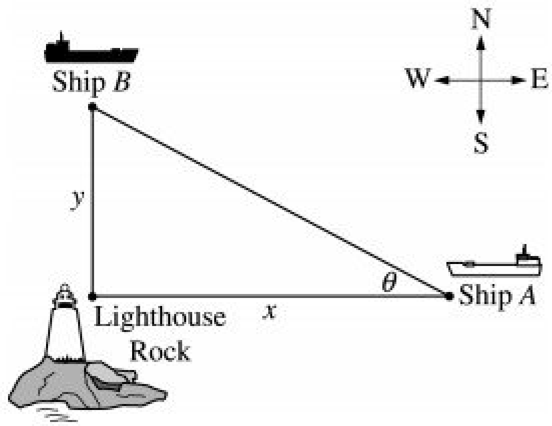
$N'(t) = S'(t) - L(t) = 0$ when $t = 5.545066$

t	$N(t)$
0	0
5.545066	1361.832842
12	195.702

The number of shoppers in the store is greatest at time $t = 5.545$ hours.

4-3-4

37.



Ship A is traveling due west toward Lighthouse Rock at a speed of 15 kilometers per hour (km/hr). Ship *B* is traveling due north away from Lighthouse rock at a speed of 10 km/hr. Let x be the distance between Ship *A* and Lighthouse Rock at time t, and let y be the distance between Ship *B* and Lighthouse Rock at time t, as shown in the figure above.

(a) Find the distance, in kilometers, between Ship *A* and Ship *B* when $x = 4$ km and $y = 3$ km.

(b) Find the rate of change, in km/hr, of the distance between the two ships when $x = 4$ km and $y = 3$ km.

(c) Let θ be the angle shown in the figure. Find the rate of change of θ, in radians per hour, when $x = 4$ km and $y = 3$ km.

Part A

1 point is earned for the answer

Distance = $\sqrt{3^2 + 4^2} = 5$ km

0	1

The student response earns all of the following points:

1 point is earned for the answer

Distance = $\sqrt{3^2 + 4^2} = 5$ km

Part B

1 point is earned for the expression for distance

2 points are earned for the differentiation with respect to t

4-3-4

<-2> chain rule error

1 point is earned for evaluation

$r^2 = x^2 + y^2$

$2r\frac{dr}{dt} = 2x\frac{dx}{dt} + 2y\frac{dy}{dt}$

or explicitly:

$r = \sqrt{x^2 + y^2}$

$\frac{dr}{dt} = \frac{1}{2\sqrt{x^2+y^2}}\left(2x\frac{dx}{dt} + 2y\frac{dy}{dt}\right)$

At $x = 4,\ y = 3,$

$\frac{dr}{dt} = \frac{4(-15)+3(10)}{5} = -6\ \text{km/hr}$

0	1	2	3	4

The student response earns all of the following points:

1 point is earned for the expression for distance

2 points are earned for the differentiation with respect to *t*

<-2> chain rule error

1 point is earned for evaluation

$r^2 = x^2 + y^2$

$2r\frac{dr}{dt} = 2x\frac{dx}{dt} + 2y\frac{dy}{dt}$

or explicitly:

$r = \sqrt{x^2 + y^2}$

$\frac{dr}{dt} = \frac{1}{2\sqrt{x^2+y^2}}\left(2x\frac{dx}{dt} + 2y\frac{dy}{dt}\right)$

At $x = 4,\ y = 3,$

$\frac{dr}{dt} = \frac{4(-15)+3(10)}{5} = -6\ \text{km/hr}$

Part C

4-3-4

1 point is earned for the expression for θ in terms of x and y

2 points are earned for differentiation with respect to t

<-2> chain rule, quotient rule, or transcendental function error

1 point is earned for the evaluation

$$\tan\theta = \frac{y}{x}$$

$$\sec^2\theta\frac{d\theta}{dt} = \frac{\frac{dy}{dt}x - \frac{dx}{dt}y}{x^2}$$

At $x = 4$ and $y = 3$, $\sec\theta = \frac{5}{4}$

$$\frac{d\theta}{dt} = \frac{16}{25}\left(\frac{10(4)-(-15)(3)}{16}\right)$$

$$= \frac{85}{25} = \frac{17}{5}\text{ radians/hr}$$

0	1	2	3	4

The student response earns all of the following points:

1 point is earned for the expression for θ in terms of x and y

2 points are earned for differentiation with respect to t

<-2> chain rule, quotient rule, or transcendental function error

1 point is earned for the evaluation

$$\tan\theta = \frac{y}{x}$$

$$\sec^2\theta\frac{d\theta}{dt} = \frac{\frac{dy}{dt}x - \frac{dx}{dt}y}{x^2}$$

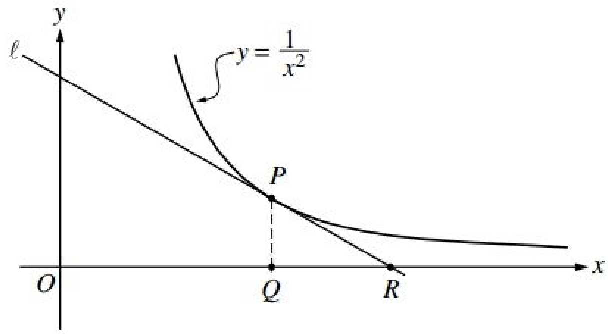
At $x = 4$ and $y = 3$, $\sec\theta = \frac{5}{4}$

$$\frac{d\theta}{dt} = \frac{16}{25}\left(\frac{10(4)-(-15)(3)}{16}\right)$$

$$= \frac{85}{25} = \frac{17}{5}\text{ radians/hr}$$

4-3-4

38.



In the figure above, line ℓ is tangent to the graph of $y = \frac{1}{x^2}$ at point P, with coordinates $\left(w, \frac{1}{w^2}\right)$, where $w > 0$. Point Q has coordinates $(w, 0)$. Line ℓ crosses the x-axis at point R, with coordinates $(k, 0)$.

(a) Find the value of k when $w = 3$.

(b) For all $w > 0$, find k in terms of w.

(c) Suppose that w is increasing at the constant rate of 7 units per second. When $w = 5$, what is the rate of change of k with respect to time?

(d) Suppose that w is increasing at the constant rate of 7 units per second. When $w = 5$, what is the rate of change of the area of n PQR with respect to time? Determine whether the area is increasing or decreasing at this instant.

Part A

1 point is earned for: $\left.\frac{dy}{dx}\right|_{x=3}$

1 point is earned for the answer

$$\frac{dy}{dx} = -\frac{2}{x^3}; \left.\frac{dy}{dx}\right|_{x=3} = -\frac{2}{27}$$

Line ℓ through $\left(3, \frac{1}{9}\right)$and $(k, 0)$ has slope $-\frac{2}{27}$.

Therefore, $\frac{0-\frac{1}{9}}{k-3} = -\frac{2}{27}$ or, $0 - \frac{1}{9} = -\frac{2}{27}(k-3)$

$k = \frac{9}{2}$

4-3-4

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for: $\left.\frac{dy}{dx}\right|_{x=3}$

1 point is earned for the answer

$$\frac{dy}{dx} = -\frac{2}{x^3}; \left.\frac{dy}{dx}\right|_{x=3} = -\frac{2}{27}$$

Line ℓ through $\left(3, \frac{1}{9}\right)$ and $(k, 0)$ has slope $-\frac{2}{27}$.

Therefore, $\frac{0-\frac{1}{9}}{k-3} = -\frac{2}{27}$ or, $0 - \frac{1}{9} = -\frac{2}{27}(k-3)$

$k = \frac{9}{2}$

Part B

1 point is earned for equation relating w and k, using slopes

1 point is earned for the answer

Line ℓ through $\left(w, \frac{1}{w^2}\right)$ and $(k, 0)$ has slope $-\frac{2}{w^3}$.

Therefore, $\frac{0-\frac{1}{w^2}}{k-w} = -\frac{2}{w^3}$ or $0 - \frac{1}{w^2} = -\frac{2}{w^3}(k-w)$

$k = \frac{3}{2}w$

4-3-4

0	1	2

The student response earns all of the following points:

1 point is earned for equation relating w and k, using slopes

1 point is earned for the answer

Line ℓ through $\left(w, \frac{1}{w^2}\right)$ and $(k, 0)$ has slope $-\frac{2}{w^3}$.

Therefore, $\frac{0-\frac{1}{w^2}}{k-w} = -\frac{2}{w^3}$ or $0 - \frac{1}{w^2} = -\frac{2}{w^3}(k - w)$

$k = \frac{3}{2}w$

Part C

1 point is earned for the answer using: $\frac{dw}{dt} = 7$

$$\frac{dk}{dt} = \frac{3}{2}\frac{dw}{dt} = \frac{3}{2} \cdot 7 = \frac{21}{2}; \left.\frac{dk}{dt}\right|_{w=5} = \frac{21}{2}$$

0	1

The student response earns all of the following points:

1 point is earned for the answer using: $\frac{dw}{dt} = 7$

$$\frac{dk}{dt} = \frac{3}{2}\frac{dw}{dt} = \frac{3}{2} \cdot 7 = \frac{21}{2}; \left.\frac{dk}{dt}\right|_{w=5} = \frac{21}{2}$$

4-3-4

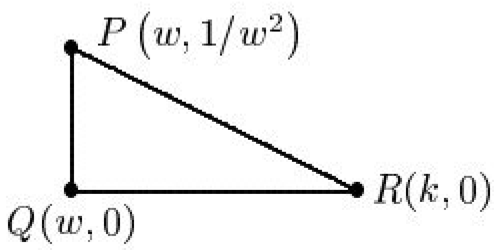
Part D

1 point is earned for the area in term or w and/or k

1 point is earned for: $\frac{dA}{dt}$ implicitly

1 point is earned for: $\left.\frac{dA}{dt}\right|_{w=5}$ using $\frac{dw}{dt} = 7$

1 point is earned for the conclusion



$A = \frac{1}{2}(k-w)\frac{1}{w^2} = \frac{1}{2}\left(\frac{3}{2}w - w\right)\frac{1}{w^2} = \frac{1}{4w}$

$\frac{dA}{dt} = -\frac{1}{4w^2}\frac{dw}{dt}$

$\left.\frac{dA}{dt}\right|_{w=5} = -\frac{1}{100} \cdot 7 = -0.07$

Therefore, area is decreasing.

0	1	2	3	4

The student response earns all of the following points:

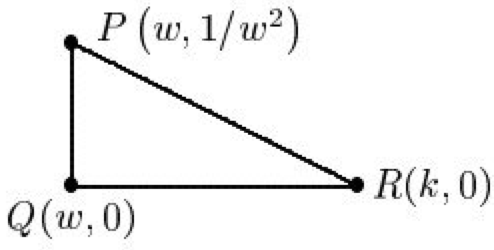
1 point is earned for the area in term or w and/or k

1 point is earned for: $\frac{dA}{dt}$ implicitly

1 point is earned for: $\left.\frac{dA}{dt}\right|_{w=5}$ using $\frac{dw}{dt} = 7$

1 point is earned for the conclusion

4-3-4



$A=\frac{1}{2}(k-w)\frac{1}{w^2}=\frac{1}{2}\left(\frac{3}{2}w-w\right)\frac{1}{w^2}=\frac{1}{4w}$

$\frac{dA}{dt}=-\frac{1}{4w^2}\frac{dw}{dt}$

$\frac{dA}{dt}\Big|_{w=5}=-\frac{1}{100}\cdot 7=-0.07$

Therefore, area is decreasing.

4-3-4

39. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (minutes)	0	3	8	12
$H(t)$ (degrees Celsius)	90	80	67	58

As a bowl of soup cools, the temperature of the soup is given by the twice-differentiable function H for $0 \leq t \leq 12$, where $H(t)$ is measured in degrees Celsius ($^\circ$C) and time t is measured in minutes. Values of $H(t)$ at selected values of time t are shown in the table above.

(a) Use the data in the table to approximate $H'(5)$. Show the computations that lead to your answer. Using correct units, explain the meaning of $H'(5)$ in the context of the problem.

(b) Is there a time t, for $0 \leq t \leq 12$, at which the temperature of the soup is 60°C ? Justify your answer.

(c) The temperature of the soup is also modeled by the twice-differentiable function C for $0 \leq t \leq 12$, where $C(t)$ is measured in degrees Celsius and time t is measured in minutes. It is known that $C(3) = 80$ and the derivative of C is given by $C'(t) = -3.6e^{-0.05t}$. Write an equation for the line tangent to the graph of C at $t = 3$ and use it to approximate $C(5)$.

(d) Based on the model given in part (c), at what rate, in degrees Celsius per minute per minute, is the rate of change of the temperature of the soup changing at time $t = 3$ minutes?

Part A

The first point requires a correct substitution of values. A simplified answer is not required.

The second point requires a time reference, rate, and units. The point may be earned with an incorrect approximation based on one computational error and provided the behavior described is consistent with the sign of the approximation, the numerical value, and the time reference.

4-3-4

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2

The student response accurately includes both of the criteria below.

- [ ] approximation
- [ ] interpretation with units

Solution:

$$H'(5) \approx \frac{H(8)-H(3)}{8-3} = \frac{67-80}{5} = -2.6$$

At time $t = 5$ minutes, the temperature of the soup is changing at a rate of -2.6 degrees Celsius per minute.

Part B

The second point may be earned if response contains conditions of IVT without naming the theorem.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2

The student response accurately includes both of the criteria below.

- [ ] H is continuous
- [ ] justification using Intermediate Value Theorem

Solution:

H is twice differentiable. $\Rightarrow H$ is continuous.

$H(12) = 58 < 60 < 90 = H(0)$

By the Intermediate Value Theorem, there is a value of t, for $0 \leq t \leq 12$, such that $H(t) = 60$.

Part C

The second point may be earned if using an incorrect value for slope, provided the value is labeled as $C'(3)$.

The third point does not require a simplified answer. Substitution of function values is required.

4-3-4

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3

The student response accurately includes all three of the criteria below.

- ☐ slope
- ☐ equation for tangent line
- ☐ approximation

Solution:

$C'(3) = -3.098549$

An equation for the tangent line is $y = 80 - 3.098549(t - 3)$.

$C(5) \approx 80 - C'(3)(5 - 3) = 73.802903$

$C(5) = 73.803$ (or 73.802)

Part D

The second point requires a time reference, rate, and units. The point may be earned with an incorrect approximation based on one computational error and provided the behavior described is consistent with the sign of the approximation, the numerical value, and the time reference.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2

The student response accurately includes both of the criteria below.

- ☐ $C''(3)$
- ☐ interpretation with units

Solution:

$C''(3) = 0.154927$

At time $t = 3$ minutes, the rate of change of the temperature is changing by 0.155 (or 0.154) degree Celsius per minute per minute.

4-3-4

40. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (hours)	0	3	8	12	14
$S'(t)$ (centimeters per hour)	2.3	2.1	1.9	1.7	1.6

The depth of snow in a field is given by the twice-differentiable function S for $0 \le t \le 14$, where $S(t)$ is measured in centimeters and time t is measured in hours. Values of $S'(t)$, the derivative of S, at selected values of time t are shown in the table above. It is known that the graph of S is concave down for $0 \le t \le 14$.

(a) Use the data in the table to approximate $S''(10)$. Show the computations that lead to your answer. Using correct units, explain the meaning of $S''(10)$ in the context of the problem.

(b) Is there a time t, for $0 \le t \le 14$, at which the depth of snow is changing at a rate of 2 centimeters per hour? Justify your answer.

(c) At time $t = 8$, the depth of snow is 45 centimeters. Use the line tangent to the graph of S at $t = 8$ to approximate the depth of snow at time $t = 10$. Is the approximation an underestimate or an overestimate of the actual depth of snow at time $t = 10$? Justify your answer.

(d) The depth of snow, in centimeters, is also modeled by the twice-differentiable function D for $0 \le t \le 14$, where $D(t) = 120 - 92e^{-\frac{t}{40}}$ and time t is measured in hours. Use the model to find $D'(10)$. Using correct units, explain the meaning of $D'(10)$ in the context of the problem.

Part A

The first point requires a correct substitution of values. A simplified answer is not required.

The second point requires a time reference, rate, and units. The point may be earned with an incorrect approximation based on one computational error and provided the behavior described is consistent with the sign of the approximation, the numerical value, and the time reference.

4-3-4

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2

The student response accurately includes both of the criteria below.

- [ ] approximation
- [ ] interpretation with units

Solution:

$$S''(10) \approx \frac{S'(12)-S'(8)}{12-8} = \frac{1.7-1.9}{4} = -0.05$$

At time $t = 10$ hours, the rate of change of the depth of snow is changing at a rate of -0.05 centimeter per hour per hour.

Part B

The second point may be earned if response contains conditions of IVT without naming the theorem.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2

The student response accurately includes both of the criteria below.

- [ ] S' is continuous
- [ ] justification using Intermediate Value Theorem

Solution:

S is twice differentiable. $\Rightarrow S'$ is differentiable and thus continuous.

$S'(8) = 1.9 < 2 < 2.1 = S'(3)$

By the Intermediate Value Theorem, there is a value of t, for $0 \leq t \leq 14$, such that $S'(t) = 2$.

Part C

The second point may be earned if using an incorrect value for slope, provided the value is labeled as $S'(8)$.

4-3-4

The second point does not require a simplified answer. Substitution of function values is required.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3

The student response accurately includes all three of the criteria below.

- [ ] equation for tangent line
- [ ] approximation
- [ ] overestimate with justification using concavity

Solution:

$S'(8) = 1.9$ from table

An equation for the tangent line is $y = 45 + 1.9(t - 8)$.

$S(10) \approx 45 + S'(8)(10 - 8) = 48.8$ centimeters

This approximation is an overestimate because the graph of S is concave down for $8 \leq t \leq 10$.

Part D

The second point requires a time reference, rate, and units. The point may be earned with an incorrect approximation based on one computational error and provided the behavior described is consistent with the sign of the approximation, the numerical value, and the time reference.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2

The student response accurately includes both of the criteria below.

- [ ] $D'(10)$
- [ ] interpretation with units

Solution:

$D'(10) = 1.791$

4-3-4

At time $t = 10$ hours, the depth of snow is changing by 1.791 centimeters per hour.

4-3-4

41. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Fish enter a lake at a rate modeled by the function E given by $E(t) = 20 + 15\sin\left(\frac{\pi t}{6}\right)$. Fish leave the lake at a rate modeled by the function L given by $L(t) = 4 + 2^{0.1t^2}$. Both $E(t)$ and $L(t)$ are measured in fish per hour, and t is measured in hours since midnight ($t = 0$).

a) How many fish enter the lake over the 5-hour period from midnight ($t = 0$) to 5 A.M. ($t = 5$) ? Give your answer to the nearest whole number.

b) What is the average number of fish that leave the lake per hour over the 5-hour period from midnight ($t = 0$) to 5 A.M. ($t = 5$) ?

c) At what time t, for $0 \leq t \leq 8$, is the greatest number of fish in the lake? Justify your answer.

d) Is the rate of change in the number of fish in the lake increasing or decreasing at 5 A.M. ($t = 5$) ? Explain your reasoning.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point requires a correct integrand and limits.

The second point requires the first point to be earned. The second point is earned with an answer of 153, 154, or any noninteger answer that is accurate to the number of places presented after the decimal point (up to the first 3). An answer between 101 and 102, inclusive, might indicate that the response uses degree mode rather than radian mode. In this case, the second point is not earned,but the remaining points for the question may be earned for consistent answers using degree mode.

4-3-4

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ integral
- ☐ answer

Solution:

$$\int_0^5 E(t)\,dt = 153.457690$$

To the nearest whole number, 153 fish enter the lake from midnight to 5 A.M.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point requires a correct integrand and limits. The $\frac{1}{5}$ is considered for the second point. If this integral is computed in Part A of the question, that number can be used in place of the integral to earn the first point.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ integral
- ☐ answer

Solution:

$$\frac{1}{5-0}\int_0^5 L(t)\,dt = 6.059038$$

The average number of fish that leave the lake per hour from midnight to 5 A.M. is 6.059 fish per hour.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

4-3-4

The first point is earned through analytical, graphical, or verbal expressions to show where $E(t) - L(t) = 0$, where $E(t) - L(t)$ changes sign, or where $E(t)$ and $L(t)$ intersect.

The second point is earned with the correct value of t.

The third point requires a global argument for the sign of $E(t) - L(t)$ rather than a local argument.

Another approach uses the Candidates Test. Let $A(t)$ be the change in the number of fish in the lake from midnight to t hours after midnight.

$$A(t) = \int_0^t (E(s) - L(s))\, ds$$

$$A'(t) = E(t) - L(t) = 0 \quad \Rightarrow \quad t = C = 6.20356$$

t	$A(t)$
0	0
C	135.01492
8	80.91908

Therefore the greatest number of fish in the lake is at time $t = 6.204$ (or 6.203).

0	1	2	3

The student response accurately includes all three of the criteria below.

- ☐ sets $E(t) - L(t) = 0$
- ☐ answer
- ☐ justification

Solution:

The rate of change in the number of fish in the lake at time t is given by $E(t) - L(t)$.

$$E(t) - L(t) = 0 \quad \Rightarrow \quad t = 6.20356$$

$E(t) - L(t) > 0$ for $0 \leq t < 6.20356$, and $E(t) - L(t) < 0$ for $6.20356 < t \leq 8$. Therefore the greatest number of fish in the lake is at time $t = 6.204$ (or 6.203).

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

4-3-4

The first point requires an attempt to use $E'(5)$ and $L'(5)$. Incorrect symbolic differentiation for both evaluated at 5 earns the point. If values are reported without symbolic work, at least one must be correct to earn the point. These values need not be reported to three places after the decimal point.

The second point requires a direct connection between $E'(5)$ and $L'(5)$. The point is not earned if reported values are incorrect. The numerical value of $E'(5) - L'(5)$ is not required, but the response must indicate that this value is negative.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ considers $E'(5)$ and $L'(5)$
- ☐ answer with explanation

Solution:

$E'(5) - L'(5) = -10.7228 < 0$

Because $E'(5) - L'(5) < 0$, the rate of change in the number of fish is decreasing at time $t = 5$.

4-3-4

42. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (days)	0	3	7	10	12
$r'(t)$ (centimeters per day)	-6.1	-5.0	-4.4	-3.8	-3.5

An ice sculpture melts in such a way that it can be modeled as a cone that maintains a conical shape as it decreases in size. The radius of the base of the cone is given by a twice-differentiable function r, where $r(t)$ is measured in centimeters and t is measured in days. The table shown gives selected values of $r'(t)$, the rate of change of the radius, over the time interval $0 \le t \le 12$.

(a) Approximate $r''(8.5)$ using the average rate of change of r' over the interval $7 \le t \le 10$. Show the computations that lead to your answer, and indicate units of measure.

(b) Is there a time t, $0 \le t \le 3$, for which $r'(t) = -6$? Justify your answer.

(c) Use a right Riemann sum with the four subintervals indicated in the table to approximate the value of $\int_0^{12} r'(t)\, dt$.

(d) The height of the cone decreases at a rate of 2 centimeters per day. At time $t = 3$ days, the radius is 100 centimeters and the height is 50 centimeters. Find the rate of change of the volume of the cone with respect to time, in cubic centimeters per day, at time $t = 3$ days. (The volume V of a cone with radius r and height h is $V = \frac{1}{3}\pi r^2 h$.)

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point the supporting work must include at least a difference and a quotient.

Simplification is not required to earn the first point. If the numerical value is simplified, it must be correct.

The second point can be earned with an incorrect approximation for $r''(8.5)$ but cannot be earned without some value for $r''(8.5)$ presented.

4-3-4

Units may be written in any equivalent form (such as $\mathrm{cm}/\mathrm{day}^2$).

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- [ ] $r''(8.5)$ with supporting work
- [ ] Units

Solution:

$r''(8.5) \approx \frac{r'(10)-r'(7)}{10-7} = \frac{-3.8-(-4.4)}{10-7}$ centimeter per day per day
$= \frac{0.6}{3} = 0.2$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, the response must establish that -6 is between $r'(0)$ and $r'(3)$ (or -6.1 and -5). This statement may be represented symbolically (with or without including one or both endpoints in an inequality) or verbally. A response of "$r'(t) = -6$ because $r'(0) = -6.1$ and $r'(3) = -5$" does not state that -6 is between -6.1 and -5. Thus this response does not earn the first point.

To earn the second point:

· The response must state that $r'(t)$ is continuous because $r'(t)$ is differentiable (or because $r(t)$ is twice differentiable).

· The response must have earned the first point. Exception: A response of "$r'(t) = -6$ because $r'(0) = -6.1$ and $r'(3) = -5$" does not earn the first point because of imprecise communication but may nonetheless earn the second point if all other criteria for the second point are met.

· The response must conclude that there is a time t such that $r'(t) = -6$ (A statement of "yes" would be sufficient.)

To earn the second point, the response need not explicitly name the Intermediate Value Theorem, but if a theorem is named, it must be correct.

0	1	2 ✓

4-3-4

The student response accurately includes **both** of the criteria below.

- ☐ $r'(0) < -6 < r'(3)$
- ☐ Conclusion using Intermediate Value Theorem

Solution:

$r(t)$ is twice-differentiable. $\Rightarrow$ $r'(t)$ is differentiable.
$\Rightarrow$ $r'(t)$ is continuous.

$r'(0) = -6.1 < -6 < -5.0 = r'(3)$

Therefore, by the Intermediate Value Theorem, there is a time t, $0 < t < 3$, such that $r'(t) = -6$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, at least seven of the eight factors in the Riemann sum must be correct. If there is any error in the Riemann sum, the response does not earn the second point.

A response of $3(-5.0) + 4(-4.4) + 3(-3.8) + 2(-3.5)$ earns both the first and second points, unless there is a subsequent error in simplification, in which case the response would earn only the first point.

A response that presents the correct answer, with accompanying work that shows the four products in the Riemann sum (without explicitly showing all of the factors and/or the sum process) does not earn the first point but earns the second point. For example, $-15 + 4(-4.4) + 3(-3.8) + -7$ does not earn the first point but earns the second point. Similarly, $-15, \ -17.6, \ -11.4, \ -7 \rightarrow -51$ does not earn the first point but earns the second point.

A response that presents the correct answer (-51) with no supporting work earns no points.

A response that provides a completely correct left Riemann sum and approximation $\int_0^{12} r'(t)\, dt$ (i.e., $3r'(0) + 4r'(3) + 3r'(7) + 2r'(10) = 3(-6.1) + 4(-5.0) + 3(-4.4) + 2(-3.8) = -59.1$) earns 1 of the 2 points.

A response that has any error in a left Riemann sum or evaluation for $\int_0^{12} r'(t)\, dt$ earns no points.

Units are not required or read in this part.

0	1	2 ✓

4-3-4

The student response accurately includes **both** of the criteria below.

- ☐ Form of right Riemann sum
- ☐ Answer

Solution:

$$\begin{aligned}\int_0^{12} r'(t)\,dt &\approx 3r'(3) + 4r'(7) + 3r'(10) + 2r'(12)\\ &= 3(-5.0) + 4(-4.4) + 3(-3.8) + 2(-3.5)\\ &= -51\end{aligned}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first 2 points could be earned in either order.

A response with a completely correct product rule, missing one or both of the correct differentials, earns the product rule point, but not the chain rule point.

For example, $\frac{dV}{dt} = \frac{2}{3}\pi rh + \frac{1}{3}\pi r^2$ earns the first point, but not the second.

A response that treats r or h (but not both) as a constant is eligible for the chain rule point but not the product rule point. For example, $\frac{dV}{dt} = \frac{2}{3}\pi rh\frac{dr}{dt}$ is correct if h is constant, and thus earns the chain rule point.

Note: Neither $\frac{dV}{dt} = \frac{2}{3}\pi r\frac{dh}{dt}$ nor $\frac{dV}{dt} = \frac{2}{3}\pi rh\frac{dr}{dt}\frac{dh}{dt}$ earns any points.

A response that assumes a functional relationship between r and h (such as $r = 2h$), and uses this relationship to create a function for volume in terms of one variable, is eligible for at most the chain rule point. For example, $r = 2h \rightarrow V = \frac{1}{3}\pi(2h)^2h = \frac{4}{3}\pi h^3 \rightarrow \frac{dV}{dt} = 4\pi h^2\frac{dh}{dt}$ earns only the chain rule point.

A response that mishandles the constant $\frac{1}{3}\pi$ cannot earn the third point but is eligible for the first 2 points.

The third point cannot be earned without both of the first 2 points.

$\frac{dV}{dt} = \frac{2}{3}\pi(100)(50)(-5) + \frac{1}{3}\pi(100)^2(-2)$ earns all 3 points.

Units are not required or read in this part.

0	1	2	3 ✓

4-3-4

The student response accurately includes **all three** of the criteria below.

- ☐ Product rule
- ☐ Chain rule
- ☐ Answer

Solution:

$$\frac{dV}{dt} = \frac{2}{3}\pi rh\frac{dr}{dt} + \frac{1}{3}\pi r^2\frac{dh}{dt}$$

$$\left.\frac{dV}{dt}\right|_{t=3} = \frac{2}{3}\pi\,(100)\,(50)\,(-5) + \frac{1}{3}\pi(100)^2\,(-2) = -\frac{70{,}000\pi}{3}$$

The rate of change of the volume of the sculpture at $t = 3$ is approximately $-\frac{70{,}000\pi}{3}$ cubic centimeters per day.

4-3-4

43. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

A constant volume of pizza dough is formed into a cylinder with a relatively small height and large radius. The dough is spun and tossed into the air in such a way that the height of the dough decreases as the radius increases, but it retains its cylindrical shape. At time $t = k$, the height of the dough is $\frac{1}{3}$ inch, the radius of the dough is 12 inches, and the radius of the dough is increasing at a rate of 2 inches per minute.

(a) At time $t = k$, at what rate is the area of the circular surface of the dough increasing with respect to time? Show the computations that lead to your answer. Indicate units of measure.

(b) At time $t = k$, at what rate is the height of the dough decreasing with respect to time? Show the computations that lead to your answer. Indicate units of measure. (The volume V of a cylinder with radius r and height h is given by $V = \pi r^2 h$.)

(c) Write an expression for the rate of change of the height of the dough with respect to the radius of the dough in terms of height h and radius r.

Part A

The first point is earned with $\frac{dA}{dt} = 2\pi r \frac{dr}{dt}$.

The second point requires substitution of values without a simplified answer.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3 ✓

The student response accurately includes all three of the criteria below.

- ☐ chain rule
- ☐ rate
- ☐ units

4-3-4

Solution:

$A = \pi r^2$

$\frac{dA}{dt} = \frac{d}{dt}\left(\pi r^2\right) = 2\pi r \frac{dr}{dt}$

At time $t = k$, $r = 12$ and $\frac{dr}{dt} = 2$.

$\frac{dA}{dt} = 2\pi \cdot 12 \cdot 2 = 48\pi$

At time $t = k$, the area of the circular surface of the dough is increasing at a rate of 48π square inches per minute.

Part B

The first and second points require evidence of product rule and chain rule and no errors. For the first 2 points, at most 1 out of 2 points is earned for partial communication of product rule and chain rule with a maximum of one computational error.

The third point requires substitution of values without a simplified answer.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3	4

The student response accurately includes all four of the criteria below.

- ☐ product rule
- ☐ chain rule
- ☐ rate
- ☐ units

Solution:

$V = \pi r^2 h$

$\frac{dV}{dt} = \frac{d}{dt}\left(\pi r^2 h\right) = 2\pi r h \frac{dr}{dt} + \pi r^2 \frac{dh}{dt}$

Because V is constant, $\frac{dV}{dt} = 0$.

At time $t = k$, $h = \frac{1}{3}$, $r = 12$, $\frac{dV}{dt} = 0$, and $\frac{dr}{dt} = 2$.

4-3-4

$$0 = 2\pi \cdot 12 \cdot \frac{1}{3} \cdot 2 + \pi \cdot 12^2 \cdot \frac{dh}{dt}$$

$$\Rightarrow \frac{dh}{dt} = -\frac{16\pi}{144\pi} = -\frac{1}{9}$$

At time $t = k$, the height of the dough is decreasing at a rate of $\frac{1}{9}$ inch per minute.

Part C

The first point requires a correct expression that demonstrates chain rule with $\frac{dV}{dr}$ in terms of r, h, and $\frac{dh}{dr}$ OR $\frac{dV}{dt}$ in terms of r,h, $\frac{dr}{dt}$, and $\frac{dh}{dt}$.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2 ✓

The student response accurately includes both of the criteria below.

- ☐ chain rule
- ☐ expression

Solution:

$$V = \pi r^2 h$$

$$\frac{dV}{dr} = \frac{d}{dr}\left(\pi r^2 h\right) = 2\pi r h + \pi r^2 \frac{dh}{dr}$$

Because V is constant, $\frac{dV}{dr} = 0$. Therefore, $0 = 2\pi r h + \pi r^2 \frac{dh}{dr} \Rightarrow \frac{dh}{dr} = \frac{-2\pi r h}{\pi r^2} = -\frac{2h}{r}$.

—OR—

$$\frac{dh}{dt} = \frac{dh}{dr} \cdot \frac{dr}{dt} \Rightarrow \frac{dh}{dr} = \frac{\frac{dh}{dt}}{\frac{dr}{dt}}$$

$$\frac{dV}{dt} = 2\pi r h \frac{dr}{dt} + \pi r^2 \frac{dh}{dt}$$

Because V is constant, $\frac{dV}{dt} = 0$. Therefore, $0 = 2\pi r h \frac{dr}{dt} + \pi r^2 \frac{dh}{dt} \Rightarrow \frac{dh}{dr} = \frac{\frac{dh}{dt}}{\frac{dr}{dt}} = \frac{-2\pi r h}{\pi r^2} = -\frac{2h}{r}$.

4-3-4

44. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

A sculptor uses a constant volume of modeling clay to form a cylinder with a large height and a relatively small radius. The clay is molded in such a way that the height of the clay increases as the radius decreases, but it retains its cylindrical shape. At time $t = c$, the height of the clay is 8 inches, the radius of the clay is 3 inches, and the radius of the clay is decreasing at a rate of $\frac{1}{2}$ inch per minute.

(a) At time $t = c$, at what rate is the area of the circular cross section of the clay decreasing with respect to time? Show the computations that lead to your answer. Indicate units of measure.

(b) At time $t = c$, at what rate is the height of the clay increasing with respect to time? Show the computations that lead to your answer. Indicate units of measure. (The volume V of a cylinder with radius r and height h is given by $V = \pi r^2 h$.)

(c) Write an expression for the rate of change of the radius of the clay with respect to the height of the clay in terms of height h and radius r.

Part A

The first point is earned with $\frac{dA}{dt} = 2\pi r \frac{dr}{dt}$.

The second point requires substitution of values without a simplified answer.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3

The student response accurately includes all three of the criteria below.

- [ ] chain rule
- [ ] rate
- [ ] units

Solution:

4-3-4

$A = \pi r^2$

$\frac{dA}{dt} = \frac{d}{dt}\left(\pi r^2\right) = 2\pi r \frac{dr}{dt}$

At time $t = c$, $r = 3$ and $\frac{dr}{dt} = -\frac{1}{2}$.

$\frac{dA}{dt} = 2\pi \cdot 3 \cdot \left(-\frac{1}{2}\right) = -3\pi$

At time $t = c$, the area of the circular surface of the dough is decreasing at a rate of 3π square inches per minute.

Part B

The first and second points require evidence of product rule and chain rule and no errors. For the first 2 points, at most 1 out of 2 points is earned for partial communication of product rule and chain rule with a maximum of one computational error.

The third point requires substitution of values without a simplified answer.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3	4 ✓

The student response accurately includes all four of the criteria below.

- ☐ product rule
- ☐ chain rule
- ☐ rate
- ☐ units

Solution:

$V = \pi r^2 h$

$\frac{dV}{dt} = \frac{d}{dt}\left(\pi r^2 h\right) = 2\pi r h \frac{dr}{dt} + \pi r^2 \frac{dh}{dt}$

Because V is constant, $\frac{dV}{dt} = 0$.

At time $t = c$, $h = 8$, $r = 3$, $\frac{dV}{dt} = 0$, and $\frac{dr}{dt} = -\frac{1}{2}$.

$0 = 2\pi \cdot 3 \cdot 8 \cdot \left(-\frac{1}{2}\right) + \pi \cdot 3^2 \cdot \frac{dh}{dt}$

4-3-4

$$\Rightarrow \frac{dh}{dt} = \frac{24\pi}{9\pi} = \frac{8}{3}$$

At time $t = c$, the height of the clay is increasing at a rate of $\frac{8}{3}$ inches per minute.

Part C

The first point requires a correct expression that demonstrates chain rule with $\frac{dV}{dh}$ in terms of r, h, and $\frac{dr}{dh}$ OR $\frac{dV}{dt}$ in terms of r, h, $\frac{dr}{dt}$, and $\frac{dh}{dt}$.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2

The student response accurately includes both of the criteria below.

- [ ] chain rule
- [ ] expression

Solution:

$V = \pi r^2 h$

$$\frac{dV}{dh} = \frac{d}{dh}\left(\pi r^2 h\right) = 2\pi rh\frac{dr}{dh} + \pi r^2$$

Because V is constant, $\frac{dV}{dh} = 0$.

Therefore, $0 = 2\pi rh\frac{dr}{dh} + \pi r^2 \quad \Rightarrow \quad \frac{dr}{dh} = \frac{-\pi r^2}{2\pi rh} = -\frac{r}{2h}$.

—OR—

$$\frac{dr}{dt} = \frac{dr}{dh} \cdot \frac{dh}{dt} \quad \Rightarrow \quad \frac{dr}{dh} = \frac{\frac{dr}{dt}}{\frac{dh}{dt}}$$

$$\frac{dV}{dt} = 2\pi rh\frac{dr}{dt} + \pi r^2\frac{dh}{dt}$$

Because V is constant, $\frac{dV}{dt} = 0$. Therefore, $0 = 2\pi rh\frac{dr}{dt} + \pi r^2\frac{dh}{dt} \quad \Rightarrow \quad \frac{dr}{dh} = \frac{\frac{dr}{dt}}{\frac{dh}{dt}} = \frac{-\pi r^2}{2\pi rh} = -\frac{r}{2h}$.

4-3-4

45. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

At an aquarium, a tank is kept full of water. At time $t = 0$, where t is measured in minutes, the water in the tank contains no salt. For time $t > 0$, water containing salt flows into the tank at a constant rate and mixes with the water in the tank. The mixed water leaves the tank through an overflow drain at the same constant rate. The number of pounds of salt in the water in the tank at time $t \geq 0$ is modeled by the function S, which satisfies the differential equation $\frac{dS}{dt} = -\frac{1}{6}(S - 36)$ with initial condition $S(0) = 0$.

(a) Is the amount of salt in the water in the tank changing more rapidly when the water contains 6 pounds of salt or when the water contains 30 pounds of salt? Explain your reasoning.

(b) Write an expression for $\frac{d^2S}{dt^2}$ in terms of S. Find the value of $\frac{d^2S}{dt^2}$ at the instant when there are 20 pounds of salt in the water in the tank. Using correct units, interpret this value in the context of the problem.

(c) Use separation of variables to find $y = S(t)$, the particular solution to the differential equation $\frac{dS}{dt} = -\frac{1}{6}(S - 36)$ with initial condition $S(0) = 0$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for an attempt to evaluate $\dfrac{dS}{dt}$ at either $S = 6$ or $S = 30$.

Correct, labeled values of $\dfrac{dS}{dt}$ at both $S = 6$ and $S = 30$ are required to earn the second point.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ Uses $\dfrac{dS}{dt}$
- ☐ Answer with reason

4-3-4

Solution:

$$\frac{dS}{dt}\bigg|_{S=6} = -\tfrac{1}{6}(6-36) = 5$$

$$\frac{dS}{dt}\bigg|_{S=30} = -\tfrac{1}{6}(30-36) = 1$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response must present $\frac{d^2S}{dt^2}$ in terms of S in order to earn the first point.

A response must present a negative value of $\frac{d^2S}{dt^2}\Big|_{S=20}$ in order to be eligible for the third point.

A response with a sign error in the second derivative (i.e., $\frac{d^2S}{dt^2} = -\frac{1}{36}(S-36)$) can earn the second point for calculating $\frac{d^2S}{dt^2}\Big|_{S=20} = -\frac{1}{36}(20-36) = \frac{4}{9}$. This response is not eligible for the third point and thus earns a maximum of 1 point in part (b).

The interpretation requires “at the moment when there are 20 pounds of salt,” “rate of the amount of salt,” “decreasing,” and “at a rate of $|\text{the presented value}|$ pounds per minute per minute” (or the equivalent) to earn the third point.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- ☐ $\frac{d^2S}{dt^2}$
- ☐ $\frac{d^2S}{dt^2}\Big|_{S=20}$
- ☐ Interpretation with units

Solution:

$$\frac{d^2S}{dt^2} = -\frac{1}{6}\frac{dS}{dt} = -\tfrac{1}{6}\left(-\tfrac{1}{6}\right)(S-36)$$

$$\frac{d^2S}{dt^2}\bigg|_{S=20} = -\tfrac{1}{6}\left(-\tfrac{1}{6}\right)(20-36) = -\tfrac{4}{9}$$

At the instant when there are 20 pounds of salt in the tank, the rate of change of the number of pounds of salt is decreasing at a rate of $\frac{4}{9}$ pound per minute per minute.

4-3-4

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response with no separation of variables earns 0 out of 4 points.

A response of $-\frac{1}{6}\int \frac{1}{S-36}\,dS = \int dt$ is a bad separation and does not earn the first point. However, this response is eligible for the second and third points. It cannot earn the fourth point.

A response that enters with both correct antiderivatives earns the first 2 points.

If the absolute value is missing on the antiderivative of $\frac{1}{S-36}$ the response earns the second point and is eligible for the third point. The response is not eligible for the fourth point.

A response with no constant of integration can earn at most the first 2 points.

A response is eligible for the third point if it has earned the first point and has at least one of the two antiderivatives correct. A response earns the third point by correctly including the constant of integration in an equation involving the presented antiderivatives and then replacing t with 0 and S with 0. This point is earned even in the presence of nonreal values (e.g., $\ln(-36)$) or severe algebraic errors.

A response is eligible for the fourth point only if it has earned the first 3 points and has correctly handled absolute value in all steps.

There are alternate (equivalent) algebraic forms of the correct equation, including, but not limited to:
$S(t) = 36 - e^{\ln 36 - t/6}$ and $S(t) = 36 - \frac{36}{e^{t/6}}$.

0	1	2	3	4

The student response accurately includes **all four** of the criteria below.

- ☐ Separation of variables
- ☐ Antiderivatives
- ☐ Constant of integration and uses initial condition
- ☐ Solves for S

Solution:

$$\frac{dS}{dt} = -\frac{1}{6}(S-36)$$

4-3-4

$$\int \frac{1}{S-36}\,dS = \int -\frac{1}{6}\,dt$$

$\ln|S-36| = -\frac{1}{6}t + C$

Because $0 \leq S < 36$, $|S-36| = 36 - S$.

$\ln(36-0) = -\frac{1}{6}(0) + C \;\Rightarrow\; \ln(36) = C$

$36 - S = 36e^{-t/6}$

$S(t) = 36 - 36e^{-t/6},\ t \geq 0$

4-3-4

46. For $0 \le t \le 31$, the rate of change of the number of mosquitoes on Tropical Island at time t days is modeled by $R(t) = 5\sqrt{t}\cos\left(\frac{t}{5}\right)$ mosquitoes per day. There are 1000 mosquitoes on Tropical Island at time $t = 0$.

(a) Show that the number of mosquitoes is increasing at time $t = 6$.

(b) At time $t = 6$, is the number of mosquitoes increasing at an increasing rate, or is the number of mosquitoes increasing at a decreasing rate? Give a reason for your answer.

(c) According to the model, how many mosquitoes will be on the island at time $t = 31$? Round your answer to the nearest whole number.

(d) To the nearest whole number, what is the maximum number of mosquitoes for $0 \le t \le 31$? Show the analysis that leads to your conclusion.

Part A

1 point is earned for showing that $R(6) > 0$

Since $R(6) = 4.438 > 0$, the number of mosquitoes is increasing at $t = 6$.

0	1 ✓

The student response earns all of the following points:

1 point is earned for showing that $R(6) > 0$

Since $R(6) = 4.438 > 0$, the number of mosquitoes is increasing at $t = 6$.

Part B

1 point is earned for considering $R'(6)$

1 point is earned for the answer with reason

$R'(6) = -1.913$

Since $R'(6) < 0$, the number of mosquitoes is increasing at a decreasing rate at $t = 6$.

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for considering $R'(6)$

4-3-4

1 point is earned for the answer with reason

$R'(6) = -1.913$

Since $R'(6) < 0$, the number of mosquitoes is increasing at a decreasing rate at $t = 6$.

Part C

1 point is earned for the integral

1 point is earned for the answer

$$1000 + \int_0^{31} R(t)dt = 964.335$$

To the nearest whole number, there are 964 mosquitoes.

0	1	2

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the answer

$$1000 + \int_0^{31} R(t)dt = 964.335$$

To the nearest whole number, there are 964 mosquitoes.

Part D

2 points are earned for absolute maximum value where 1 point is earned for the integral and 1 point is earned for the answer

2 points are earned for analysis where 1 point is earned for computing interior critical points and 1 point is earned for completing analysis

$R(t) = 0$ when $t = 0$, $t = 2.5\pi$, or $t = 7.5\pi$

$R(t) > 0$ on $0 < t < 2.5\pi$

$R(t) < 0$ on $2.5\pi < t < 7.5\pi$

$R(t) > 0$ on $7.5\pi < t < 31$

4-3-4

The absolute maximum number of mosquitoes occurs at $t = 2.5\pi$ or at $t = 31$.

$$1000 + \int_0^{2.5\pi} R(t)dt = 1039.357,$$

There are 964 mosquitoes at $t = 31$, so the maximum number of mosquitoes is 1039, to the nearest whole number.

0	1	2	3	4

The student response earns all of the following points:

2 points are earned for absolute maximum value where 1 point is earned for the integral and 1 point is earned for the answer

2 points are earned for analysis where 1 point is earned for computing interior critical points and 1 point is earned for completing analysis

$R(t) = 0$ when $t = 0$, $t = 2.5\pi$, or $t = 7.5\pi$

$R(t) > 0$ on $0 < t < 2.5\pi$

$R(t) < 0$ on $2.5\pi < t < 7.5\pi$

$R(t) > 0$ on $7.5\pi < t < 31$

The absolute maximum number of mosquitoes occurs at $t = 2.5\pi$ or at $t = 31$.

$$1000 + \int_0^{2.5\pi} R(t)dt = 1039.357,$$

There are 964 mosquitoes at $t = 31$, so the maximum number of mosquitoes is 1039, to the nearest whole number.

4-3-4

47. The penguin population on an island is modeled by a differentiable function P of time t, where $P(t)$ is the number of penguins and t is measured in years, for $0 \le t \le 40$. There are 100,000 penguins on the island at time $t = 0$. The birth rate for the penguins on the island is modeled by

$$B(t) = 1000e^{0.06t} \text{ penguins per year}$$

and the death rate for the penguins on the island is modeled by

$$D(t) = 250e^{0.1t} \text{ penguins per year.}$$

(a) What is the rate of change of the penguin population on the island at time $t = 0$?

(b) To the nearest whole number, what is the penguin population on the island at time $t = 40$?

(c) To the nearest whole number, what is the average rate of change of the penguin population on the island for $0 \le t \le 40$?

(d) To the nearest whole number, find the absolute minimum penguin population and the absolute maximum penguin population on the island for $0 \le t \le 40$. Show the analysis that leads to your answer.

Part A

The response can earn up to 1 point:

1 point is earned for the answer

$P'(0) = B(0) - D(0) = 1000 - 250 = 750$ penguins per year

0	1

The response earns all of the following point:

1 point is earned for the answer

$P'(0) = B(0) - D(0) = 1000 - 250 = 750$ penguins per year

Part B

The response can earn up to 2 points:

1 point is earned for correct limits

1 point is earned for the integrand

1 point is earned for answer

$$P(40) = 100000 + \int_0^{40} (B(t) - D(t))\, dt$$

$= 100000 + 33057.56459$

4-3-4

There are 133,058 penguins on the island.

0	1	2	3

The student earns all of the following point:

1 point is earned for correct limits

1 point is earned for the integrand

1 point is earned for answer

$P(40) = 100000 + \int_0^{40} (B(t) - D(t))\, dt$

$= 100000 + 33057.56459$

There are 133,058 penguins on the island.

Part C

The response can earn up to 1 point:

1 point is earned for the answer

$\frac{1}{40}\int_0^{40} (B(t) - D(t))\, dt = 826.439$

OR

$\frac{P(40) - P(0)}{40 - 0} = \frac{133058 - 100000}{40} = 826.45$

The average rate of change is 826 penguins per year.

0	1

The student earns all of the following point:

1 point is earned for the answer

$\frac{1}{40}\int_0^{40} (B(t) - D(t))\, dt = 826.439$

4-3-4

OR

$$\frac{P(40) - P(0)}{40 - 0} = \frac{133058 - 100000}{40} = 826.45$$

The average rate of change is 826 penguins per year.

Part D

The response can earn up to 4 points:

1 point is earned for the expression $B(t)$ - $D(t) = 0$

$B(t)$ - $D(t) = 0$

1 point is earned for the solution for t

$$1000e^{0.06t} = 250e^{0.1t} \Rightarrow t = A = \frac{\ln 4}{0.04} = 34.657359$$

The absolute minimum and absolute maximum occur at a critical point or at an endpoint.

1 point is earned for the minimum value

1 point is earned for the maximum value

$P(0) = 100000$

$$P(A) = 100000 + \int_0^A (B(t) - D(t))\, dt = 139166.667$$

$P(40) = 133058$

The minimum population is 100,000 and the maximum population is 139,167 penguins.

0	1	2	3	4

The student earns all of the following point:

1 point is earned for the expression $B(t)$ - $D(t) = 0$

$B(t)$ - $D(t) = 0$

1 point is earned for the solution for t

$$1000e^{0.06t} = 250e^{0.1t} \Rightarrow t = A = \frac{\ln 4}{0.04} = 34.657359$$

4-3-4

The absolute minimum and absolute maximum occur at a critical point or at an endpoint.

1 point is earned for the minimum value

1 point is earned for the maximum value

$P(0) = 100000$

$P(A) = 100000 + \int_0^A (B(t) - D(t))\, dt = 139166.667$

$P(40) = 133058$

The minimum population is 100,000 and the maximum population is 139,167 penguins.

48. For $1 \leq t \leq 7$ hours, people enter a store on a certain day at a rate modeled by $E(t) = 45 + 24\sin t$ and leave at a rate modeled by $L(t) = 10 + 7e^{0.21t}$, where $E(t)$ and $L(t)$ are measured in people per hour. During the time period $1 \leq t \leq 7$, on which of the following intervals is the number of people in the store decreasing?

(A) $1.487 \leq t \leq 4.884$

(B) $4.033 \leq t \leq 5.788$ ✓

(C) $1 \leq t \leq 3.167$ and $6.236 \leq t \leq 7$

(D) $1 \leq t \leq 4.033$ and $5.788 \leq t \leq 7$

Answer B

Correct. The rate of change of the number of people in the store is $E(t) - L(t)$. The number of people is decreasing on intervals where this rate of change is non-positive, that is, where $E(t) - L(t) \leq 0$. The graphing calculator is used to solve $E(t) - L(t) = 0$ and examine the graph of $E(t) - L(t)$. Since $E(t) - L(t) = 0$ at $t = 4.033$ and $t = 5.788$, the graph of $E(t) - L(t)$ indicates that $E(t) - L(t)$ is non-positive for $4.033 \leq t \leq 5.788$.

4-3-4

49. In a certain factory, assume that the number of workers is constant. The number of minutes N that it takes to make a single unit of a product and the number of units U of the product that are made per day satisfy the relationship $U = \frac{k}{N}$, where k is a constant. Which of the following best describes the relationship between the rate of change, with respect to time t, of U and the rate of change, with respect to time t, of N ?

(A) $\frac{dU}{dt} = \frac{k}{\left(\frac{dN}{dt}\right)}$

(B) $\frac{dU}{dt} = \frac{-k}{\left(\frac{dN}{dt}\right)}$

(C) $\frac{dU}{dt} = \frac{k}{N^2}\left(\frac{dN}{dt}\right)$

(D) $\frac{dU}{dt} = \frac{-k}{N^2}\left(\frac{dN}{dt}\right)$ ✓

Answer D

Correct. Differentiating both sides of $U = \frac{k}{N}$ with respect to t, and using the fact that k is a constant, gives this equation. Using the power rule, the derivative of $\frac{1}{N} = N^{-1}$ is $-N^{-2} = \frac{-1}{N^2}$ and using the chain rule, $\frac{dU}{dt} = \left(\frac{d}{dN}\left(\frac{k}{N}\right)\right)\left(\frac{dN}{dt}\right) = \frac{-k}{N^2}\left(\frac{dN}{dt}\right)$.

50. The radius of a right circular cylinder is increasing at a rate of 2 units per second. The height of the cylinder is decreasing at a rate of 5 units per second. Which of the following expressions gives the rate at which the volume of the cylinder is changing with respect to time in terms of the radius r and height h of the cylinder?
(The volume V of a cylinder with radius r and height h is $V = \pi r^2 h$.)

(A) $-20\pi r$

(B) $2\pi r h$

(C) $4\pi r h - 5\pi r^2$ ✓

(D) $4\pi r h + 5\pi r^2$

Answer C

$\frac{dV}{dt} = 2\pi r h \frac{dr}{dt} + \pi r^2 \frac{dh}{dt}$, where $\frac{dr}{dt} = 2$ and $\frac{dh}{dt} = -5$

4-3-4

51. Let x and y be functions of time t such that the sum of x and twice y is constant. Which of the following equations describes the relationship between the rate of change of x with respect to time and the rate of change of y with respect to time?

(A) $\frac{dx}{dt} = 2\frac{dy}{dt}$

(B) $\frac{dx}{dt} = -2\frac{dy}{dt}$ ✓

(C) $2\frac{dx}{dt} + \frac{dy}{dt} = 0$

(D) $\frac{dx}{dt} + 2\frac{dy}{dt} = K$, where K is a function of t

Answer B

Correct. It is given that $x + 2y = K$, where K is a constant. Differentiating both sides of the equation with respect to t and using the fact that the derivative of a constant is 0 gives $\frac{dx}{dt} + 2\frac{dy}{dt} = 0$, which implies that $\frac{dx}{dt} = -2\frac{dy}{dt}$.

52. A right triangle has base x meters and height h meters, where h is constant and x changes with respect to time t, measured in seconds. The angle θ, measured in radians, is defined by $\tan\theta = \frac{h}{x}$. Which of the following best describes the relationship between $\frac{d\theta}{dt}$, the rate of change of θ with respect to time, and $\frac{dx}{dt}$, the rate of change of x with respect to time?

(A) $\frac{d\theta}{dt} = \left(\frac{-h}{x^2+h^2}\right)\frac{dx}{dt}$ radians per second ✓

(B) $\frac{d\theta}{dt} = \left(\frac{h}{x^2+h^2}\right)\frac{dx}{dt}$ radians per second

(C) $\frac{d\theta}{dt} = \left(\frac{-h}{x\sqrt{x^2+h^2}}\right)\frac{dx}{dt}$ radians per second

(D) $\frac{d\theta}{dt} = \left(\frac{h}{x\sqrt{x^2+h^2}}\right)\frac{dx}{dt}$ radians per second

Answer A

Correct. Differentiating both sides of the equation with respect to t and using the chain rule on the left side gives $\frac{d}{dt}(\tan\theta) = \frac{d}{dt}\left(\frac{h}{x}\right) \Rightarrow \sec^2\theta\frac{d\theta}{dt} = \frac{-h}{x^2}\frac{dx}{dt}$. Since $\tan\theta = \frac{h}{x}$ implies that $\sec\theta = \frac{\sqrt{x^2+h^2}}{x}$, substituting for $\sec\theta$ and solving for $\frac{d\theta}{dt}$ yields the answer.

4-3-4

53. Charles's law states that if the pressure of a dry gas is held constant, then the volume V of the gas and its temperature T, measured in degrees Kelvin, satisfy the relationship $V = cT$, where c is a constant. Which of the following best describes the relationship between the rate of change, with respect to time t, of the volume and the rate of change, with respect to time t, of the temperature?

(A) $\frac{dV}{dt} = T\frac{dc}{dt}$

(B) $\frac{dV}{dt} = c\frac{dT}{dt}$ ✓

(C) $\frac{dV}{dT} = c$

(D) $1 = c\frac{dT}{dV}$

Answer B

Correct. Differentiating both sides of $V = cT$ with respect to t and using the fact that c is a constant, gives this equation. In this context, t and T represent different quantities, and t is the independent variable. $\frac{dT}{dt}$ results from application of the chain rule with the derivative of T.

54. A rectangle has width w inches and height h inches, where the width is twice the height. Both w and h are functions of time t, measured in seconds. If A represents the area of the rectangle, which of the following gives the rate of change of A with respect to t ?

(A) $\frac{dA}{dt} = 4h \text{ in/sec}$

(B) $\frac{dA}{dt} = 3h\frac{dh}{dt} \text{ in}^2\text{/sec}$

(C) $\frac{dA}{dt} = 4h\frac{dh}{dt} \text{ in/sec}$

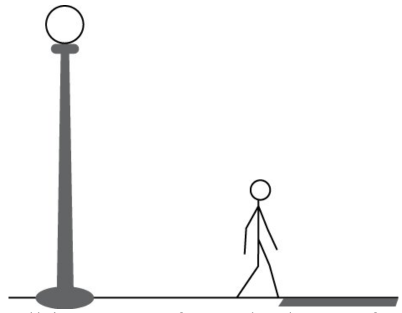
(D) $\frac{dA}{dt} = 4h\frac{dh}{dt} \text{ in}^2\text{/sec}$ ✓

Answer D

Correct. Given that the width is twice the height, then $A = wh = \left(2h\right)h = 2h^2$. Differentiating with respect to time and using the chain rule gives $\frac{dA}{dt} = 4h\frac{dh}{dt}$. Using the units for h and $\frac{dh}{dt}$, the units for $\frac{dA}{dt}$ are $\text{in} \cdot \left(\text{in/sec}\right)$ or $\text{in}^2\text{/sec}$.

4-3-4

55.



A person whose height is M feet is walking away from the base of a streetlight along a straight road, as shown in the figure above. The height of the streetlight is L feet. At time t seconds, the person is x feet from the streetlight, and the length of the person's shadow is s feet. The quantities are related by the equation $\frac{1}{L}(x+s)=\frac{1}{M}s$, where L and M are constants. Which of the following best describes the relationship between the rate of change of x with respect to time and the rate of change of s with respect to time?

(A) $\frac{dx}{dt}=\frac{L}{M}s-s$

(B) $\frac{dx}{dt}=\frac{L}{M}s-\frac{ds}{dt}$

(C) $\frac{dx}{dt}=\frac{L}{M}\frac{ds}{dt}-s$

(D) $\frac{dx}{dt}=\frac{L}{M}\frac{ds}{dt}-\frac{ds}{dt}$

Answer D

Correct. Differentiating both sides of the given equation with respect to time yields $\frac{1}{L}\left(\frac{dx}{dt}+\frac{ds}{dt}\right)=\frac{1}{M}\frac{ds}{dt}$. Solving for $\frac{dx}{dt}$ gives $\frac{dx}{dt}=\frac{L}{M}\frac{ds}{dt}-\frac{ds}{dt}$.

4-3-4

56. Let x and y be functions of time t such that the sum of x and y is constant. Which of the following equations describes the relationship between the rate of change of x with respect to time and the rate of change of y with respect to time?

(A) $\frac{dx}{dt} = \frac{dy}{dt}$

(B) $\frac{dx}{dt} = -\frac{dy}{dt}$ ✓

(C) $\frac{dx}{dt} + \frac{dy}{dt} = \frac{dK}{dt}$, where K is a function of t

(D) $\frac{dx}{dt} + \frac{dy}{dt} = K$, where K is a function of t

Answer B

Correct. It is given that $x + y = K$, where K is a constant. Differentiating both sides of the equation with respect to t and using the fact that the derivative of a constant is 0 gives $\frac{dx}{dt} + \frac{dy}{dt} = 0$, which implies that $\frac{dx}{dt} = -\frac{dy}{dt}$.

57. Boyle's law states that if the temperature of an ideal gas is held constant, then the pressure P of the gas and its volume V satisfy the relationship $P = \frac{k}{V}$, where k is a constant. Which of the following best describes the relationship between the rate of change, with respect to time t, of the pressure and the rate of change, with respect to time t, of the volume?

(A) $\frac{dP}{dt} = \frac{k}{\left(\frac{dV}{dt}\right)}$

(B) $\frac{dP}{dt} = \frac{-k}{\left(\frac{dV}{dt}\right)}$

(C) $\frac{dP}{dt} = \frac{k}{V^2}\left(\frac{dV}{dt}\right)$

(D) $\frac{dP}{dt} = \frac{-k}{V^2}\left(\frac{dV}{dt}\right)$ ✓

Answer D

Correct. Differentiating both sides of $P = \frac{k}{V}$ with respect to t, and using the fact that k is a constant, gives this equation. Using the power rule, the derivative of $\frac{1}{V} = V^{-1}$ is $-V^{-2} = \frac{-1}{V^2}$ and using the chain rule, $\frac{dP}{dt} = \left[\frac{d}{dV}\left(\frac{k}{V}\right)\right]\left(\frac{dV}{dt}\right) = \frac{-k}{V^2}\left(\frac{dV}{dt}\right)$.

4-3-4

58. A right triangle has base x feet and height h feet, where x is constant and h changes with respect to time t, measured in seconds. The angle θ, measured in radians, is defined by $\tan\theta = \frac{h}{x}$. Which of the following best describes the relationship between $\frac{d\theta}{dt}$, the rate of change of θ with respect to time, and $\frac{dh}{dt}$, the rate of change of h with respect to time?

(A) $\frac{d\theta}{dt} = \left(\frac{x}{x^2+h^2}\right)\frac{dh}{dt}$ radians per second ✓

(B) $\frac{d\theta}{dt} = \left(\frac{x^2}{x^2+h^2}\right)\frac{dh}{dt}$ radians per second

(C) $\frac{d\theta}{dt} = \left(\frac{1}{\sqrt{x^2+h^2}}\right)\frac{dh}{dt}$ radians per second

(D) $\frac{d\theta}{dt} = \tan^{-1}\left(\frac{1}{x}\frac{dh}{dt}\right)$ radians per second

Answer A

Correct. Differentiating both sides of the equation with respect to t and using the chain rule on the left-hand side, gives $\frac{d}{dt}(\tan\theta) = \frac{d}{dt}\left(\frac{h}{x}\right) \Rightarrow \sec^2\theta\frac{d\theta}{dt} = \frac{1}{x}\frac{dh}{dt}$. Since $\tan\theta = \frac{h}{x}$ implies that $\sec\theta = \frac{\sqrt{x^2+h^2}}{x}$, substituting for $\sec\theta$ and solving for $\frac{d\theta}{dt}$ yields the answer.

59. Ohm's law states that if the resistance of the path of current between two points is constant, then the voltage difference V between the points and the current I flowing between the points, measured in amperes, satisfy the relationship $V = cI$, where c is a constant. Which of the following best describes the relationship between the rate of change with respect to time t of the voltage and the rate of change with respect to time t of the current?

(A) $\frac{dV}{dt} = I\frac{dc}{dt}$

(B) $\frac{dV}{dt} = c\frac{dI}{dt}$ ✓

(C) $\frac{dV}{dI} = c$

(D) $1 = c\frac{dI}{dV}$

Answer B

Correct. Differentiating both sides of $V = cI$ with respect to t and using the fact that c is a constant gives this equation. In this context, t and I represent different quantities, and t is the independent variable. $c\frac{dI}{dt}$ results from application of the chain rule in taking the derivative of cI.

4-3-4

60. A triangle has base b centimeters and height h centimeters, where the height is three times the base. Both b and h are functions of time t, measured in seconds. If A represents the area of the triangle, which of the following gives the rate of change of A with respect to t ?

(A) $\frac{dA}{dt} = 3b \text{ cm/sec}$

(B) $\frac{dA}{dt} = 2b\frac{db}{dt} \text{ cm}^2\text{/sec}$

(C) $\frac{dA}{dt} = 3b\frac{db}{dt} \text{ cm/sec}$

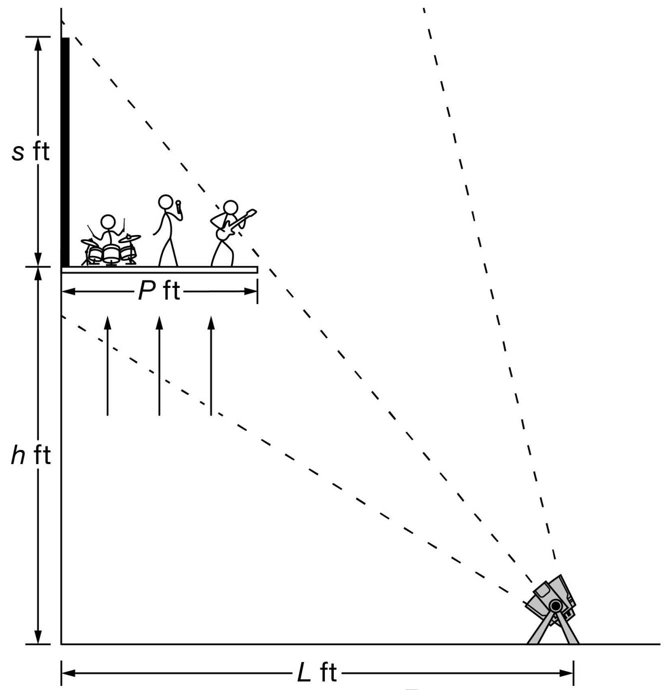
(D) $\frac{dA}{dt} = 3b\frac{db}{dt} \text{ cm}^2\text{/sec}$

Answer D

Correct. Given that the height is three times the base, then $A = \frac{bh}{2} = \frac{b(3b)}{2} = \frac{3}{2}b^2$. Differentiating with respect to time and using the chain rule gives $\frac{dA}{dt} = 3b\frac{db}{dt}$. Using the units for b and $\frac{db}{dt}$, the units for $\frac{dA}{dt}$ are $(\text{cm})\,(\text{cm/sec})$ or $\text{cm}^2\text{/sec}$.

4-3-4

61.



At a concert, a band is playing on a platform that extends P feet from a wall behind the band, and the platform is rising from ground level, as shown in the figure above. A light source is L feet from the wall, and the platform casts a lengthening shadow on the wall as the platform rises. At time t seconds, the platform is h feet above the ground, and the height of the shadow is s feet. The quantities are related by the equation $\frac{1}{L}(h+s)=\frac{1}{P}s$, where L and P are constants. Which of the following best expresses the rate of change of h with respect to time in terms of the rate of change of s with respect to time?

(A) $\frac{dh}{dt}=\frac{L}{P}s-s$

(B) $\frac{dh}{dt}=\frac{L}{P}s-\frac{ds}{dt}$

(C) $\frac{dh}{dt}=\frac{L}{P}\frac{ds}{dt}-s$

(D) $\frac{dh}{dt}=\frac{L}{P}\frac{ds}{dt}-\frac{ds}{dt}$

Answer D

Correct. Differentiating both sides of the given equation with respect to time yields $\frac{1}{L}\left(\frac{dh}{dt}+\frac{ds}{dt}\right)=\frac{1}{P}\frac{ds}{dt}$. Solving for $\frac{dh}{dt}$ gives $\frac{dh}{dt}=\frac{L}{P}\frac{ds}{dt}-\frac{ds}{dt}$.

4-3-4

62. On a certain workday, the rate, in tons per hour, at which unprocessed gravel arrives at a gravel processing plant is modeled by $G(t) = 90 + 45\cos\left(\frac{t^2}{18}\right)$, where t is measured in hours and $0 \le t \le 8$. At the beginning of the workday ($t = 0$), the plant has 500 tons of unprocessed gravel. During the hours of operation, $0 \le t \le 8$, the plant processes gravel at a constant rate of 100 tons per hour.

(a) Find $G'(5)$. Using correct units, interpret your answer in the context of the problem.

(b) Find the total amount of unprocessed gravel that arrives at the plant during the hours of operation on this workday.

(c) Is the amount of unprocessed gravel at the plant increasing or decreasing at time $t = 5$ hours? Show the work that leads to your answer.

(d) What is the maximum amount of unprocessed gravel at the plant during the hours of operation on this workday? Justify your answer.

Part A

1 point is earned for $G'(5)$

1 point is earned for the interpretation with units

$G'(5) = -24.588$ (or -24.587)

The rate at which gravel is arriving is decreasing by 24.588 (or 24.587) tons per hour per hour at time $t = 5$ hours.

0	1	2

The student response earns all of the following points:

1 point is earned for $G'(5)$

1 point is earned for the interpretation with units

$G'(5) = -24.588$ (or -24.587)

The rate at which gravel is arriving is decreasing by 24.588 (or 24.587) tons per hour per hour at time $t = 5$ hours.

Part B

1 point is earned for the integral

1 point is earned for the answer

$$\int_0^8 G(t)\,dt = 825.551 \text{ tons}$$

4-3-4

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the answer

$$\int_0^8 G(t)\,dt = 825.551 \text{ tons}$$

Part C

1 point is earned for comparing $G(5)$ to 100

1 point is earned for the conclusion

$G(5) = 98.140764 < 100$

At time $t = 5$, the rate at which unprocessed gravel is arriving is less than the rate at which it is being processed.

Therefore, the amount of unprocessed gravel at the plant is decreasing at time $t = 5$.

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for comparing $G(5)$ to 100

1 point is earned for the conclusion

$G(5) = 98.140764 < 100$

At time $t = 5$, the rate at which unprocessed gravel is arriving is less than the rate at which it is being processed.

Therefore, the amount of unprocessed gravel at the plant is decreasing at time $t = 5$.

Part D

1 point is earned for considering $A'(t) = 0$

1 point is earned for the answer

4-3-4

1 point is earned for the justification

The amount of unprocessed gravel at time t is given by $A\left(t\right)=500+\int_0^t\left(G\left(s\right)-100\right)ds.$

$A'(t)=G(t)-100=0 \Rightarrow t=4.923480$

t	$A(t)$
0	500
4.92348	635.376123
8	525.551089

The maximum amount of unprocessed gravel at the plant during this workday is 635.376 tons.

0	1	2	3

The student response earns all of the following points:

1 point is earned for considering $A'(t)=0$

1 point is earned for the answer

1 point is earned for the justification

The amount of unprocessed gravel at time t is given by $A\left(t\right)=500+\int_0^t\left(G\left(s\right)-100\right)ds.$

$A'(t)=G(t)-100=0 \Rightarrow t=4.923480$

t	$A(t)$
0	500
4.92348	635.376123
8	525.551089

The maximum amount of unprocessed gravel at the plant during this workday is 635.376 tons.

4-3-4

63. At time t, $t \geq 0$, the volume of a sphere is increasing at a rate proportional to the reciprocal of its radius. At $t = 0$, the radius of the sphere is 1 and at $t = 15$, the radius is 2. (The volume V of a sphere with a radius r is $V = 4/3\pi r^3$.)

(a) Find the radius of the sphere as a function of t.

(b) At what time t will the volume of the sphere be 27 *times* its volume at $t = 0$?

Part A

1 point is earned for $\frac{dV}{dt} = \frac{k}{r}$

1 point is earned for $\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$

1 point is earned for separates variables

1 point is earned for integration (must have C)

1 point is earned for solving for C

1 point is earned for solving for k

1 point is earned for answer

$\frac{dV}{dt} = \frac{k}{r}$
$\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$
$\frac{k}{r} = 4\pi r^2 \frac{dr}{dt}$
$kdt = 4\pi r^3 dr \int kdt = \int 4\pi r^3 \frac{dr}{dt} dt$
$kt + C = \pi r^4$
At $t = 0,\ r = 1,$ so $C = \pi$
At $t = 15,\ r = 2,$ so
$15k + \pi = 16\pi, k = \pi$
$\pi r^4 = \pi t + \pi$
$r = \sqrt[4]{t+1}$

0	1	2	3	4	5	6	7

4-3-4

The student response earns all of the following points:

1 point is earned for $\dfrac{dV}{dt} = \dfrac{k}{r}$

1 point is earned for $\dfrac{dV}{dt} = 4\pi r^2 \dfrac{dr}{dt}$

1 point is earned for separates variables

1 point is earned for integration (must have C)

1 point is earned for solving for C

1 point is earned for solving for k

1 point is earned for answer

$\frac{dV}{dt} = \frac{k}{r}$
$\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$
$\frac{k}{r} = 4\pi r^2 \frac{dr}{dt}$
$kdt = 4\pi r^3 dr \int kdt = \int 4\pi r^3 \frac{dr}{dt} dt$
$kt + C = \pi r^4$
At $t = 0,\ r = 1,$ so $C = \pi$
At $t = 15,\ r = 2,$ so
$15k + \pi = 16\pi, k = \pi$
$\pi r^4 = \pi t + \pi$
$r = \sqrt[4]{t+1}$

Part B

1 point is earned for $r = 3$

1 point is earned for answer

At $t = 0$, $r = 1$, so $V(0) = \dfrac{4}{3}\pi$

4-3-4

$27V(0) = 27\left(\frac{4}{3}\pi\right) = 36\pi$

$36\pi = \frac{4}{3}\pi r^3$

$r = 3$

$\sqrt[4]{t+1} = 3$

$t = 80$

0	1	2

The student response earns all of the following points:

1 point is earned for $r = 3$

1 point is earned for answer

At $t = 0$, $r = 1$, so $V(0) = \dfrac{4}{3}\pi$

$27V(0) = 27\left(\frac{4}{3}\pi\right) = 36\pi$

$36\pi = \frac{4}{3}\pi r^3$

$r = 3$

$\sqrt[4]{t+1} = 3$

$t = 80$

4-3-4

64. The radius r of a sphere is increasing at a constant rate of 0.04 centimeters per second. (*Note*: The volume of a sphere with radius r is $V = 4/3\pi r^3$.)

(a) At the time when the radius of the sphere is 10 centimeters, what is the rate of increase of its volume?

(b) At the time when the volume of the sphere is $36\ \pi$ cubic centimeters, what is the rate of increase of the area of a cross section through the center of the sphere?

(c) At the time when the volume and the radius of the sphere are increasing at the same numerical rate, what is the radius?

Part A

2 points are earned for derivative

2 points are **deducted** for chain rule error

1 point is **deducted** for constant error

1 point is earned for evaluation

$$\frac{dV}{dt} = \frac{4}{3} \cdot 3\pi r^2 \frac{dr}{dt}$$

$\therefore$ when $r = 10$, $\dfrac{dr}{dt} = 0.04$

$$\frac{dV}{dt} = 4\pi 10^2 (0.04) = \boxed{16\pi \text{ cm}^3/\text{sec}}$$

0	1	2	3

The student response earns all of the following points:

2 points are earned for derivative

2 points are **deducted** for chain rule error

1 point is **deducted** for constant error

1 point is earned for evaluation

$$\frac{dV}{dt} = \frac{4}{3} \cdot 3\pi r^2 \frac{dr}{dt}$$

$\therefore$ when $r = 10$, $\dfrac{dr}{dt} = 0.04$

4-3-4

$$\frac{dV}{dt} = 4\pi 10^2 (0.04) = \boxed{16\pi \text{ cm}^3/\text{sec}}$$

Part B

1 point is earned for finding $r = 3$

2 points are earned for derivative

1 point is **deducted** for incorrect area

1 point is **deducted** for each constant error

1 point is earned for evaluation

$$V = 36\pi \Rightarrow 36 = \frac{4}{3}r^3 \Rightarrow r^3 = 27 \Rightarrow r = 3$$

$$A = \pi r^2$$
$$\frac{dA}{dt} = 2\pi r\frac{dr}{dt}$$

$\therefore$ when $V = 36\pi$, $\dfrac{dr}{dt} = .04$

$$\frac{dA}{dt} = 2\pi \cdot 3 (\cdot 04) = \frac{6\pi}{25} = \boxed{\cdot 24\pi \text{ cm}^2/\text{sec}}$$

0	1	2	3	4

The student response earns all of the following points:

1 point is earned for finding $r = 3$

2 points are earned for derivative

1 point is **deducted** for incorrect area

1 point is **deducted** for each constant error

4-3-4

1 point is earned for evaluation

$V = 36\pi \Rightarrow 36 = \frac{4}{3}r^3 \Rightarrow r^3 = 27 \Rightarrow r = 3$

$A = \pi r^2$

$\frac{dA}{dt} = 2\pi r \frac{dr}{dt}$

$\therefore$ when $V = 36\pi$, $\frac{dr}{dt} = .04$

$\frac{dA}{dt} = 2\pi \cdot 3\,(\cdot 04) = \frac{6\pi}{25} = \boxed{\cdot 24\pi \text{ cm}^2/\text{sec}}$

Part C

1 point is earned if sets $\frac{dV}{dt} = \frac{dr}{dt}$

1 point is earned for solution to equation

$\frac{dV}{dt} = \frac{dr}{dt}$

$4\pi r^2 \frac{dr}{dt} = \frac{dr}{dt} \Rightarrow 4\pi r^2 = 1$

$\therefore r^2 = \frac{1}{4\pi} \Rightarrow \boxed{r = \frac{1}{2\sqrt{\pi}} \text{ cm}}$

0	1	2

The student response earns all of the following points:

1 point is earned if sets $\frac{dV}{dt} = \frac{dr}{dt}$

4-3-4

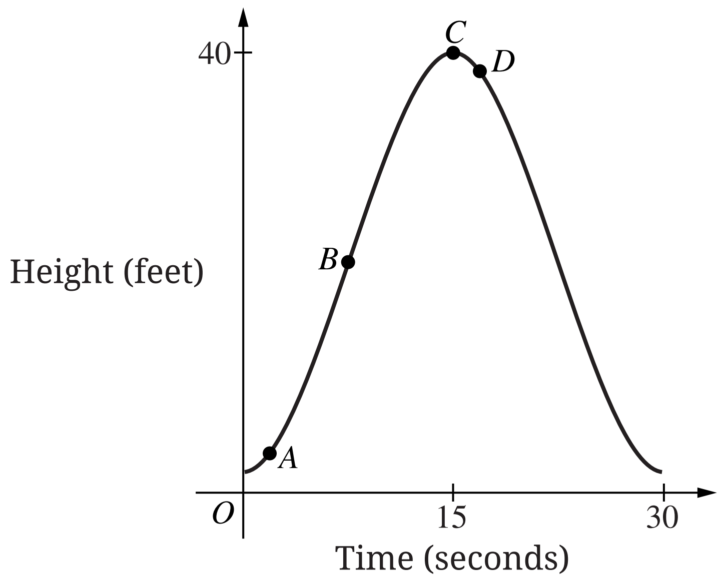
1 point is earned for solution to equation

$$\frac{dV}{dt} = \frac{dr}{dt}$$

$$4\pi r^2 \frac{dr}{dt} = \frac{dr}{dt} \Rightarrow 4\pi r^2 = 1$$

$$\therefore r^2 = \frac{1}{4\pi} \Rightarrow \boxed{r = \frac{1}{2\sqrt{\pi}} \text{ cm}}$$

65.



An amusement park ride reaches a maximum height of 40 feet above the ground. The graph shows the height above the ground, in feet, of a passenger on the ride for times $0 \leq t \leq 30$ seconds. At which of the following points is the height of the passenger above the ground changing the fastest?

(A) A

(B) B

(C) C

(D) D

Answer B

Correct. The rate at which the height is changing is given by the derivative of the height. The rate is changing the fastest where the derivative is a maximum. This occurs where the graph of the height has an inflection point, which is point B.

4-3-4

66. The tide removes sand from Sandy Point Beach at a rate modeled by the function R, given by

$$R(t) = 2 + 5\sin\left(\frac{4\pi t}{25}\right).$$

A pumping station adds sand to the beach at a rate modeled by the function S, given by

$$S(t) = \frac{15t}{1+3t}.$$

Both $R(t)$ and $S(t)$ have units of cubic yards per hour and t is measured in hours for $0 \leq t \leq 6$. At time $t =$ 0, the beach contains 2500 cubic yards of sand.

(a) How much sand will the tide remove from the beach during this 6-hour period? Indicate units of measure.

(b) Write an expression for $Y(t)$, the total number of cubic yards of sand on the beach at time t.

(c) Find the rate at which the total amount of sand on the beach is changing at time $t = 4$.

(d) For $0 \leq t \leq 6$, at what time t is the amount of sand on the beach a minimum? What is the minimum value? Justify your answers.

Part A

1 point is earned for integral

1 point is earned for the answer with units

$$\int_0^6 R(t)dt = 31.815 \text{ or } 31.816 \text{ yd}^3$$

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for integral

1 point is earned for the answer with units

$$\int_0^6 R(t)dt = 31.815 \text{ or } 31.816 \text{ yd}^3$$

Part B

1 point is earned for the integrand

1 point is earned for the limits

4-3-4

1 point is earned for the answer

$$Y(t) = 2500 + \int_0^t (S(x) - R(x))dx$$

0	1	2	3

The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the limits

1 point is earned for the answer

$$Y(t) = 2500 + \int_0^t (S(x) - R(x))dx$$

Part C

1 point is earned for the answer

$$Y'(t) = S(t) - R(t)$$
$$Y'(4) = S(4) - R(4) = -1.908 \text{ or } -1.909 \text{ yd}^3/\text{hr}$$

0	1

The student response earns all of the following points:

1 point is earned for the answer

$$Y'(t) = S(t) - R(t)$$
$$Y'(4) = S(4) - R(4) = -1.908 \text{ or } -1.909 \text{ yd}^3/\text{hr}$$

Part D

1 point is earned for setting $Y'(t) = 0$

1 point is earned for the critical t-value

4-3-4

1 point is earned for the answer with justification

$Y'(t) = 0$ when $S(t) - R(t) = 0$.

The only value in [0, 6] to satisfy $S(t) = R(t)$ is $a = 5.117865$.

t	$Y(t)$
0	2500
a	2492.3694
6	2493.2766

The amount of sand is a minimum when $t = 5.117$ or 5.118 hours. The minimum value is 2492.369 cubic yards.

0	1	2	3

The student response earns all of the following points:

1 point is earned for setting $Y'(t) = 0$

1 point is earned for the critical t-value

1 point is earned for the answer with justification

$Y'(t) = 0$ when $S(t) - R(t) = 0$.

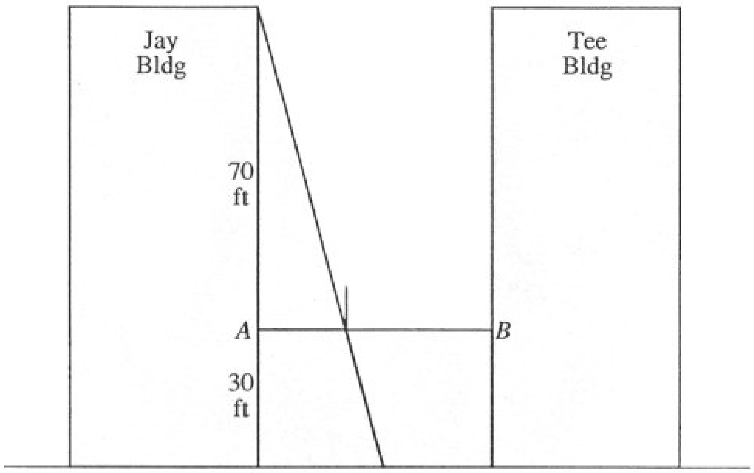
The only value in [0, 6] to satisfy $S(t) = R(t)$ is $a = 5.117865$.

t	$Y(t)$
0	2500
a	2492.3694
6	2493.2766

The amount of sand is a minimum when $t = 5.117$ or 5.118 hours. The minimum value is 2492.369 cubic yards.

4-3-4

67. A tight rope is stretched 30 feet above the ground between the Jay and the Tee buildings, which are 50 feet apart. A tightrope walker, walking at a constant rate of 2 feet per second from point A to point B, is illuminated by a spotlight 70 feet above point A, as shown in the diagram.



(a) How fast is the shadow of the tightrope walker's feet moving along the ground when she is midway between the buildings? (Indicate units of measure.)

(b) How far from point A is the tightrope walker when the shadow of her feet reaches the base of the Tee Building? (Indicate units of measure.)

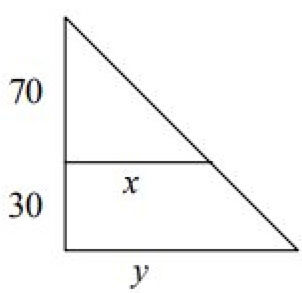
(c) How fast is the shadow of the tightrope walker's feet moving up the wall of the Tee Building when she is 10 feet from point B? (Indicate units of measure.)

Part A

1 point is earned for establishing two-variable equation relating variables "x" and "y"

1 point is earned for establishing equation relating rates dx/dt, dy/dt

1 point is earned for finding answer



$$\frac{y}{100} = \frac{x}{70}$$

4-3-4

$$70\frac{dy}{dt} = 100\frac{dx}{dt} \text{ or } \frac{dy/dt}{100} = \frac{dx/dt}{70}$$
$$70\frac{dy}{dt} = 100(2)$$
$$\frac{dy}{dt} = \frac{10}{7}(2)$$
$$\boxed{\frac{20}{7}\text{ ft/sec}}$$

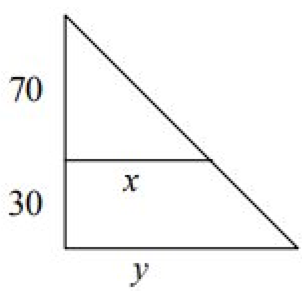
0	1	2	3

The student response earns all of the following points:

1 point is earned for establishing two-variable equation relating variables "x" and "y"

1 point is earned for establishing equation relating rates dx/dt, dy/dt

1 point is earned for finding answer



$$\frac{y}{100} = \frac{x}{70}$$

$$70\frac{dy}{dt} = 100\frac{dx}{dt} \text{ or } \frac{dy/dt}{100} = \frac{dx/dt}{70}$$
$$70\frac{dy}{dt} = 100(2)$$
$$\frac{dy}{dt} = \frac{10}{7}(2)$$
$$\boxed{\frac{20}{7}\text{ ft/sec}}$$

Part B

2 points are earned for derived correctly

4-3-4

2 points are **deducted** for incorrect proportion

2 points are **deducted** for value other than 50 used for *y*

1 point is **deducted** for arithmetic error

$$\frac{x}{70} = \frac{50}{100}$$
$$\boxed{x = 35 \text{ ft}}$$

0	1	2

The student response earns all of the following points:

2 points are earned for derived correctly

2 points are **deducted** for incorrect proportion

2 points are **deducted** for value other than 50 used for *y*

1 point is **deducted** for arithmetic error

$$\frac{x}{70} = \frac{50}{100}$$
$$\boxed{x = 35 \text{ ft}}$$

Part C

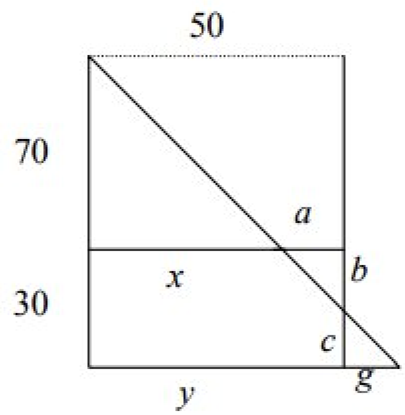
1 point is earned for establishing proportion relating horizontal variables x, a, or g to vertical variables b or c

2 points are earned for establishing equation relating rates from student's proportion

Max 1/2 if student stops before getting an equation involving at most two variables and their rates

1 point is earned for answer (including units)

4-3-4



$\frac{b}{a} = \frac{70}{x}$ OR

$\frac{b}{50-x} = \frac{70}{x}$

$bx = 3500 - 70x$

$b\frac{dx}{dt} + x\frac{db}{dt} = -70\frac{dx}{dt}$

$\left(\frac{35}{2}\right)(2) + (40)\frac{db}{dt} = -70(2)$

$\frac{db}{dt} = -\frac{35}{8}$

$\boxed{\frac{35}{8}\ \text{ft/sec.}}$

0	1	2	3	4

The student response earns all of the following points:

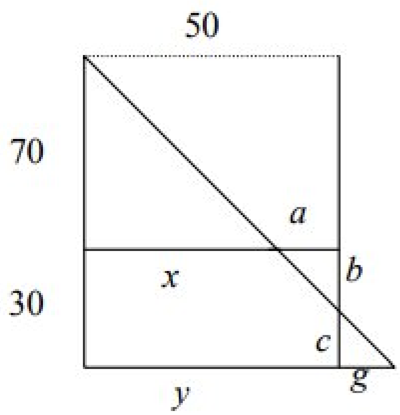
1 point is earned for establishing proportion relating horizontal variables x, a, or g to vertical variables b or c

2 points are earned for establishing equation relating rates from student's proportion

Max 1/2 if student stops before getting an equation involving at most two variables and their rates

1 point is earned for answer (including units)

4-3-4



$\frac{b}{a} = \frac{70}{x}$ OR

$\frac{b}{50-x} = \frac{70}{x}$

$bx = 3500 - 70x$

$b\frac{dx}{dt} + x\frac{db}{dt} = -70\frac{dx}{dt}$

$\left(\frac{35}{2}\right)(2) + (40)\frac{db}{dt} = -70(2)$

$\frac{db}{dt} = -\frac{35}{8}$

$\boxed{\frac{35}{8} \text{ ft/sec.}}$